

Healthcare, Heart Failure

Complete technical documentation of a 19-worksheet Tableau survival profiling dashboard, extracted from the workbook XML

DASHBOARD	Dashboard 1
WORKBOOK	Healthcare - Heart Failure.twb (packaged workbook)
AUTHOR	Atharva Devne
PROJECT TYPE	Self-directed, for the author's own skill development
DATASET	Kaggle, public dataset. heart_failure_clinical_records_dataset.csv, 13 columns
TABLEAU	Desktop 2025.1.0 (20251.25.0313.2002), macOS, doc format 18.1
CANVAS	1900 x 1050 px, sizing-mode='range', black, fully tiled
SCALE	19 worksheets, 1 dashboard, 13 calculated fields, 1 action
STATUS	Published to Tableau Public, revision 1.2

SCREENSHOT SLOT

Paste your own dashboard screenshot here. No real screenshot could be captured: no Tableau runtime was available and the network policy denied public.tableau.com.

TABLEAU PUBLIC LINK

Recoverable from the XML, to be verified by you: workbook id Healthcare-HeartFailure_17429983482570, dashboard repository id Dashboard1, current revision 1.2.

Dashboard 1 - Healthcare - Heart Failure (1900 x 1050 px, range sizing)				
Coordinate-accurate wireframe reconstructed from <zones> in the .twb XML. Not a screenshot.				
Sheet 15 KPI: Total Individuals	Sheet 15 (2) KPI: Total Deaths	Sheet 15 (3) KPI: Total Males	Sheet 15 (4) KPI: Total Females	Sheet 15 (5) KPI: Average Age
Sheet 1 Anaemia-Survival Stats	Sheet 1 (3) High Blood Pressure	Sheet 1 (5) Smoking	Sheet 1 (2) Diabetes	Sheet 1 (4) Sex
Sheet 6 Age histogram	Sheet 6 (2) Creatinine Phosphokinase Filter: Sex (dropdown)	Sheet 6 (9) Age-Sex scatter	Sheet 6 (3) Ejection Fraction Color legend: Death Event	Sheet 6 (4) Platelets
Sheet 6 (6) Serum Sodium	Sheet 6 (7) Time	Sheet 6 (8) Age vs Time scatter		Sheet 6 (5) Serum Creatinine

Coordinate-accurate wireframe reconstructed from the <zones> geometry in the .twb XML. Not a screenshot.

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PART A

Full technical documentation

Sections 1 to 19: overview, data sources, data model, preparation, field inventory, calculated fields, parameters, filters, KPIs, worksheet analysis, visualization breakdown, layout, actions, tooltips, business logic, enabled insights, technical summary, interview explanation and resume bullets.

1. DASHBOARD OVERVIEW

Item	Value	Evidence
Dashboard name	Dashboard 1	✓ DIRECTLY VERIFIED (<dashboard name='Dashboard 1'>)
Workbook name	Healthcare - Heart Failure.twb (inside a packaged archive)	✓ DIRECTLY VERIFIED (archive member name)
Tableau version	Document format version='18.1', original-version='18.1'; source-build='2025.1.0 (20251.25.0313.2002)'; build comment 20243.25.0110.1701	✓ DIRECTLY VERIFIED
Source platform	source-platform='mac'	✓ DIRECTLY VERIFIED
Publish target	repository-location derived-from='https://public.tableau.com/workbooks/Healthcare-HeartFailure_17429983482570?rev=1.1', id='Healthcare-HeartFailure_17429983482570', revision='1.2'	✓ DIRECTLY VERIFIED
xml:base	https://haproxy-traffic-splitter	✓ DIRECTLY VERIFIED
Author / owner	Atharva Devne. Confirmed by the author. The workbook itself stores no author field.	✓ CONFIRMED BY AUTHOR (not from the file)
Author evidence in the file	The source file path is /Users/atharvadevne/UIC/Business Data Visualization/Heart Fail Prediction, so the macOS account was atharvadevne. The folder name is where the file happened to sit, not a course deliverable.	✓ VERIFIED path; context ✓ CONFIRMED BY AUTHOR
Dashboard dimensions	1900 x 1050 px, sizing-mode='range' (min and max both 1900 x 1050)	✓ DIRECTLY VERIFIED
Dashboard background	#000000 (black)	✓ DIRECTLY VERIFIED

Item	Value	Evidence
Dashboard UUID	{14818E80-7948-4984-90AD-D1E566C79BE9}	✓ DIRECTLY VERIFIED

Dashboards contained in the workbook

#	Dashboard name	Analyzed here
1	Dashboard 1	Yes. It is the only dashboard in the workbook.

Stories: none. There is no `<stories>` element. ✓ DIRECTLY VERIFIED

Purpose, business problem, audience, story

These four items are **not stored as text anywhere in the workbook**. There is no dashboard description, no caption, no annotation, and no text object. ✗ NOT AVAILABLE as stated fact.

What can be read off the structure (△ INFERRED, from worksheet titles, encodings and filters):

- **Purpose (inferred).** Every single chart on the dashboard encodes `Death Event` on Color, and 12 of the 14 chart titles end with the literal words `- Survival Stats`. The dashboard is a *survival profiling* view: for each clinical attribute in the dataset, show how the population splits between `Survive` and `Death`.
- **Business / clinical question (inferred).** "Among heart failure patients, how does each clinical and demographic attribute relate to mortality during the follow-up period?"
- **Project type (confirmed by the author). Self-directed. Built for the author's own skill development**, not for a course, a client or an employer. The `UIC/Business Data Visualization` folder in the path is simply where the file sat.
- **Dataset origin (confirmed by the author). Kaggle.** It is a **public** dataset, which is why embedding a full extract and publishing to Tableau Public raises no disclosure concern.
- **Intended audience (inferred).** The content (clinical biomarkers, comorbidity flags, follow-up time) is framed for a clinical analytics or population health reader. In practice the audience is anyone reviewing the author's portfolio.
- **Overall story (inferred).** Top row states the cohort size and its composition. Middle rows split each binary comorbidity by outcome. Lower rows show the distribution of each continuous biomarker by outcome, ending with two scatter views that bring age, follow-up time and sex together.

2. DATA SOURCE INVENTORY

There is exactly **one** data source in this workbook.

Data Source	Type	Connection	Tables/Files	Important Fields	Live/Extract	Notes
heart_failure_clinical_records_dataset (internal name federated.15n4u4511e3xfk1749m9r1i714e1)	Federated wrapper over a flat file	class='textscan', directory='/Users/atharvadevne/UIC/Business Data Visualization/Heart Fail Prediction', filename='heart_failure_clinical_records_dataset.csv', server='', workgroup-auth-mode='as-is'	1 table: [heart_failure_clinical_records_data set#csv]	age, DEATH_EVENT, ejection_fraction, serum_creatinine, serum_sodium, platelets, creatinine_phosphokinase, time, sex, smoking, diabetes, anaemia, high_blood_pressure	Extract , enabled='true', count='-1', units='records'	count='-1' means all rows are extracted, no sampling. The extract is materialised in the packaged archive as Data/tableau-temp/#TableauTemp_1yyvr7n0uz5nmh12wm62y1t1ntc5.hyper

✓ DIRECTLY VERIFIED for all cells above.

Detailed data source attributes

Attribute	Value	Evidence
Data source caption	heart_failure_clinical_records_dataset	✓
Internal name	federated.15n4u4511e3xfk1749m9r1i714e1	✓
inline	true (embedded in the workbook, not a published data source)	✓
Named connection	textscan.0eo7kao1bn80dv0zsqdco0t9bxql	✓
Connection class	textscan (Tableau's delimited text file connector)	✓
File name	heart_failure_clinical_records_dataset.csv	✓
Directory	/Users/atharvadevne/UIC/Business Data Visualization/Heart Fail Prediction	✓
Database / schema	None. A flat CSV has no database or schema.	✓ (absence)
Number of tables	1	✓
Text parsing options	character-set='UTF-8', header='yes', locale='en_US', separator=','	✓
Extract name	Not stored as a friendly name. The physical file is #TableauTemp_1yyvr7n0uz5nmh12wm62y1t1ntc5.hyper	✓
Extract row limit	count='-1', units='records' (no limit)	✓
Extract is user-specific	false	✓
Data source filters	None. No <filter> exists at the <datasource> level.	✓ (absence)
Extract filters	None. The <extract> element carries no filter child.	✓ (absence)
Custom SQL	None. The relation type='table', not type='text'. No SQL text anywhere.	✓ (absence)
Aliases	Yes , <aliases enabled='yes'>. Six fields carry value aliases (see section 4).	✓
Renamed fields	Yes , 13 of 13 source columns carry a caption that differs from the raw column name.	✓
Hidden fields	None. No column carries hidden='true'.	✓ (absence)
Geographic roles	None. No semantic-role or geographic role assignment exists. No map is used.	✓ (absence)
Data source calculations	13 calculated columns (7 bins, 5 string labels, 1 constant). See section 6.	✓

3. DATA MODEL / RELATIONSHIPS / JOINS

There is no data model to speak of. This is a single-table data source. ✓ DIRECTLY VERIFIED

Check	Result
Joins	None. Only one <code><relation type='table'></code> exists.
Relationships (noodles)	None.
Unions	None.
Blending	None. Only one data source exists, so blending is impossible.
Nested joins	None.
Cardinality settings	Not applicable, nothing to relate.
Referential integrity settings	Not applicable.
Custom SQL	None.
Join filters	None.

Plain English

The workbook reads one CSV file, `heart_failure_clinical_records_dataset.csv`, as a single flat table of 13 columns. Tableau wraps it in a `federated` connection (that is just Tableau's standard container since version 2020.2, it does not imply federation across systems here). The whole table is then materialised into a Hyper extract. Every worksheet and the dashboard all point at that one table. No second table is ever introduced, so there is nothing to join, relate, union or blend.

Text representation

```
heart_failure_clinical_records_dataset.csv  (13 columns, 1 table)
|
|  full extract, count = -1 (all records), units = records
v
#TableauTemp_1yyvr7n0uz5nmh12wm62y1t1ntc5.hyper
|
v
federated.15n4u4511e3xfk1749m9r1i714e1
  "heart_failure_clinical_records_dataset"
  |
  +--> 19 worksheets
  +--> Dashboard 1
```

4. DATA PREPARATION / CLEANING

Every transformation listed here is present in the workbook XML. Nothing is claimed that is not.

4.1 Field renaming (captions)

All 13 raw CSV columns were given display captions. The raw snake_case names are preserved as the field identity, the caption is what appears in the UI.

Raw column (name)	Caption applied	Affected object
age	Age	measure
anaemia	Anaemia	dimension
creatinine_phosphokinase	Creatinine Phosphokinase	measure
diabetes	Diabetes	dimension
DEATH_EVENT	Death Event	dimension
ejection_fraction	Ejection Fraction	measure
high_blood_pressure	High Blood Pressure	dimension
platelets	Platelets	measure
serum_creatinine	Serum Creatinine	measure
serum_sodium	Serum Sodium	measure
sex	Sex	dimension
smoking	Smoking	dimension
time	Time	measure

What was done: raw machine column names were replaced with human-readable labels. **Why it was likely necessary (△ inferred):** the CSV uses lowercase snake_case and one all-caps name, which reads badly in axis titles, legends and tooltips.

4.2 Role conversion (measure to dimension)

Six integer columns were converted from Tableau's default `measure` role to `dimension` , with `type='ordinal'` .

Field	New role	New type	Default aggregation retained
[DEATH_EVENT]	dimension	ordinal	Sum
[anaemia]	dimension	ordinal	Sum
[diabetes]	dimension	ordinal	Sum

Field	New role	New type	Default aggregation retained
[high_blood_pressure]	dimension	ordinal	Sum
[sex]	dimension	ordinal	Sum
[smoking]	dimension	ordinal	Sum

What was done: the 0/1 flag columns were reclassified as categorical dimensions. **Why it was likely necessary (△ inferred):** Tableau imports 0 / 1 integers as measures and would default to SUM(). Summing a flag is meaningless for these fields. Making them dimensions lets them drive Color, Label and Filter shelves.

The seven genuinely continuous columns were left as measures: age , creatinine_phosphokinase , ejection_fraction , platelets , serum_creatinine , serum_sodium , time .

4.3 Value aliasing (0/1 decoding)

<aliases enabled='yes'> at the data source level, with six aliased fields. This is the single most substantive cleaning step in the workbook.

Field	Key 0 becomes	Key 1 becomes	Note
[DEATH_EVENT]	Survive	Death	The string is literally Survive in the XML. It is a typo for "Survive" and it will render that way on the dashboard.
[anaemia]	Negative	Positive	
[diabetes]	Negative	Positive	
[high_blood_pressure]	Negative	Positive	
[sex]	Female	Male	
[smoking]	Non-Smoker	Smoker	The value has a leading space: " Non-Smoker" .

What was done: raw binary codes were decoded to clinical language at the data source layer, so every worksheet inherits the decoding. **Why it was likely necessary (△ inferred):** a legend reading "0 / 1" is unreadable; "Survive / Death" and "Positive / Negative" are directly interpretable by a clinical reader. **Two defects worth flagging:** the Survive spelling and the leading space in " Non-Smoker" . Both are real values in the file and both are user-visible.

4.4 Binning

Seven bins were created, one for each continuous measure. All are <calculation class='bin'> with peg='0' (bins anchored at zero) and a fixed numeric size .

Bin field	Source field	Bin size	decimals
[Age (bin)]	[age]	4.48	0

Bin field	Source field	Bin size	decimals
[Creatinine Phosphokinase (bin)]	[creatinine_phosphokinase]	376	2
[Ejection Fraction (bin)]	[ejection_fraction]	8.22	0
[Platelets (bin)]	[platelets]	41767	4
[Serum Creatinine (bin)]	[serum_creatinine]	0.772	-1
[Serum Sodium (bin)]	[serum_sodium]	3.57	0
[Time (bin)]	[time]	15.1	1

What was done: each continuous biomarker was discretised into fixed-width buckets. **Why it was likely necessary (△ inferred):** these bin sizes are not round numbers. 4.48 , 8.22 , 0.772 , 3.57 , 15.1 are the values Tableau proposes by default when you right-click a measure and choose Create > Bins, because Tableau suggests a size derived from the field's range and row count. **The sizes were accepted as offered rather than hand-set.** That is a defensible choice, but it is worth knowing if you are asked about it in an interview.

4.5 Transformations that are NOT present

Stated explicitly so nothing is over-claimed:

Transformation	Present?
Null handling (ISNULL , IFNULL , ZN)	No. No such function appears in any formula.
Duplicate handling / dedup logic	No.
Data type conversions (INT() , STR() , FLOAT() , DATE())	No. All types came from the CSV scan.
Date conversions or date parts	No. The dataset has no date field at all. time is an integer follow-up period, not a date.
String manipulation (LEFT , SPLIT , TRIM , REPLACE , CONTAINS)	No.
Split fields	No.
Pivot	No.
Union	No.
Groups	No. No <group> element exists.
Sets	No.
Hierarchies / drill paths	No.
Geographic role assignment	No.
Tableau Prep flow	No. Nothing in the archive references Prep.
Data source filter	No.
Extract filter	No.

5. COMPLETE FIELD INVENTORY

5.1 Source fields (13, from the CSV)

Field Name (caption)	Original Name	Data Type	Role	Aggregation	Table	Description / Usage
Age	age	real	Measure	Sum (default); used as AVG and CNT in views	heart_failure_clinical_reco rds_dataset.csv	Drives the Age histogram, the Average Age KPI, the Age-Sex box scatter and the Age-Time scatter
Anaemia	anaemia	integer	Dimension (ordinal)	Sum	same	Color + Angle + Label on the Anaemia donut
Creatinine Phosphokinase	creatinine_phos phokinase	integer	Measure	Sum	same	Binned, drives the CPK histogram
Death Event	DEATH_EVENT	integer	Dimension (ordinal)	Sum	same	The outcome variable. On Color in every chart, and it is the field the highlight action brushes on
Diabetes	diabetes	integer	Dimension (ordinal)	Sum	same	Color + Angle + Label on the Diabetes donut
Ejection Fraction	ejection_fracti on	integer	Measure	Sum	same	Binned, drives the Ejection Fraction histogram
High Blood Pressure	high_blood_pres sure	integer	Dimension (ordinal)	Sum	same	Color + Angle + Label on the HBP donut
Platelets	platelets	real	Measure	Sum	same	Binned, drives the Platelets histogram
Serum Creatinine	serum_creatinin e	real	Measure	Sum	same	Binned, drives the Serum Creatinine histogram
Serum Sodium	serum_sodium	integer	Measure	Sum	same	Binned, drives the Serum Sodium histogram
Sex	sex	integer	Dimension (ordinal)	Sum	same	Donut dimension, KPI filter for Total Males / Total Females, dashboard filter control, and column split on the Age-Sex scatter
Smoking	smoking	integer	Dimension (ordinal)	Sum	same	Color + Angle + Label on the Smoking donut
Time	time	integer	Measure	Sum	same	Binned for the Time histogram; used raw on the Age-Time scatter

5.2 Calculated fields (13)

Field Name	Class	Data Type	Role	Formula
[Age (bin)]	bin	integer	Dimension (quantitative)	[age] , size 4.48
[Creatinine Phosphokinase (bin)]	bin	integer	Dimension (quantitative)	[creatinine_phosphokinase] , size 376
[Ejection Fraction (bin)]	bin	integer	Dimension (quantitative)	[ejection_fraction] , size 8.22

Field Name	Class	Data Type	Role	Formula
[Platelets (bin)]	bin	integer	Dimension (quantitative)	[platelets] , size 41767
[Serum Creatinine (bin)]	bin	integer	Dimension (quantitative)	[serum_creatinine] , size 0.772
[Serum Sodium (bin)]	bin	integer	Dimension (quantitative)	[serum_sodium] , size 3.57
[Time (bin)]	bin	integer	Dimension (quantitative)	[time] , size 15.1
Total Individuals ([Calculation_215609893383794688])	tableau	string	Dimension (nominal)	"Total Individuals"
Total Deaths ([Calculation_215609893384032257])	tableau	string	Dimension (nominal)	"Total Deaths"
Total Males ([Calculation_215609893384187906])	tableau	string	Dimension (nominal)	"Total Males"
Total Females ([Calculation_215609893384642563])	tableau	string	Dimension (nominal)	"Total Females"
Average Age ([Calculation_215609893384794116])	tableau	string	Dimension (nominal)	"Average Age"
0 ([Calculation_144326355385667585])	tableau	integer	Measure	0

5.3 Field categorisation

Category	Members
Dimensions (from source)	Anaemia, Death Event, Diabetes, High Blood Pressure, Sex, Smoking
Measures (from source)	Age, Creatinine Phosphokinase, Ejection Fraction, Platelets, Serum Creatinine, Serum Sodium, Time
Dates	None. The dataset contains no date or datetime field.
Geographic fields	None.
Calculated fields	The 13 listed in 5.2
Parameters	None. There is no Parameters data source and no param-domain-type column anywhere in this workbook.
Sets	None.
Groups	None.
Bins	7 (listed in 5.2)

5.4 Usage classification

Classification	Fields
Used on the dashboard (every worksheet placed on Dashboard 1 uses them)	All 13 source fields, all 7 bins, all 5 label calcs, and the 0 calc
Used only in worksheets, not surfaced as a control	All except Sex, which additionally appears as a dashboard filter control, and Death Event, which additionally appears as a dashboard color legend
Used inside other calculations	age, creatinine_phosphokinase, ejection_fraction, platelets, serum_creatinine, serum_sodium, time (each is the input to its bin)
Hidden	None

Classification	Fields
Unused	None. Every declared field is referenced by at least one worksheet.

6. CALCULATED FIELDS

6.1 The seven bin calculations

All seven follow the same pattern, so one worked example plus the table is enough.

Calculated Field: Age (bin)

- **Exact Tableau definition:** `<calculation class='bin' decimals='0' formula='[age]' peg='0' size='4.48' />`
- **Data type:** integer
- **Role / type:** dimension, quantitative, `aggregation='None'`
- **Referenced fields:** `[age]`
- **Used in worksheet(s):** Sheet 6 (on the Columns shelf)
- **Used in KPI(s):** none
- **Purpose:** turn a continuous age measure into fixed-width buckets so a count can be drawn as a histogram.
- **Plain English:** every patient is placed into an age bucket 4.48 years wide, with the first bucket anchored at 0.
- **Business logic:** age is a risk factor, and a histogram of counts split by outcome shows where in the age range deaths concentrate.
- **Dependencies:** none. It depends only on a raw source column.

Bin calculated field	formula	size	peg	decimals	Used in worksheet
Age (bin)	<code>[age]</code>	4.48	0	0	Sheet 6
Creatinine Phosphokinase (bin)	<code>[creatinine_phosphokinase]</code>	376	0	2	Sheet 6 (2)
Ejection Fraction (bin)	<code>[ejection_fraction]</code>	8.22	0	0	Sheet 6 (3)
Platelets (bin)	<code>[platelets]</code>	41767	0	4	Sheet 6 (4)
Serum Creatinine (bin)	<code>[serum_creatinine]</code>	0.772	0	-1	Sheet 6 (5)
Serum Sodium (bin)	<code>[serum_sodium]</code>	3.57	0	0	Sheet 6 (6)
Time (bin)	<code>[time]</code>	15.1	0	1	Sheet 6 (7)

6.2 The five KPI label calculations

Calculated Field: **Total Individuals**

- **Exact Tableau formula:** "Total Individuals"
- **Internal name:** [Calculation_215609893383794688]
- **Data type:** string
- **Role / type:** dimension, nominal
- **Referenced fields:** none. It is a string literal.
- **Used in worksheet(s):** Sheet 15
- **Used in KPI(s):** the Total Individuals KPI card
- **Purpose:** supply the caption text that sits next to the number on the KPI card.
- **Plain English:** a constant text field that always returns the words "Total Individuals".
- **Business logic:** none, it is presentation only.
- **Dependencies:** none.

The same pattern for the other four:

Calculated field	Internal name	Exact formula	Used in worksheet
Total Individuals	[Calculation_215609893383794688]	"Total Individuals"	Sheet 15
Total Deaths	[Calculation_215609893384032257]	"Total Deaths"	Sheet 15 (2)
Total Males	[Calculation_215609893384187906]	"Total Males"	Sheet 15 (3)
Total Females	[Calculation_215609893384642563]	"Total Females"	Sheet 15 (4)
Average Age	[Calculation_215609893384794116]	"Average Age"	Sheet 15 (5)

Technique note (△ inferred): placing a constant string on the Text shelf next to a measure is a common way to build a KPI card without using a title or a text object, because the caption then inherits the worksheet's font formatting and moves with the mark.

6.3 The axis placeholder calculation

Calculated Field: **0**

- **Exact Tableau formula:** 0
- **Internal name:** [Calculation_144326355385667585]
- **Data type:** integer
- **Role:** measure

- **Scope:** declared with `user:unnamed='Sheet 1'`, that is, it was created ad hoc inside `Sheet 1` and then promoted to the data source
- **Referenced fields:** none
- **Used in worksheet(s):** all five donut sheets: `Sheet 1`, `Sheet 1 (2)`, `Sheet 1 (3)`, `Sheet 1 (4)`, `Sheet 1 (5)`
- **Used in KPI(s):** none
- **Purpose:** give the pie chart a numeric axis so that a **dual axis** can be created, which is what turns a pie into a donut.
- **Exact shelf expression:** the Rows shelf of each donut sheet is literally `([federated.15n4u4511e3xfk1749m9r1i714e1].[sum:Calculation_144326355385667585:qk] + [federated.15n4u4511e3xfk1749m9r1i714e1].[sum:Calculation_144326355385667585:qk])` which is `SUM([0])` placed on Rows twice, the standard dual-axis construction.
- **Plain English:** a measure that is always zero. Two copies of it create two overlapping axes at the same position. The first axis carries the coloured pie, the second carries a smaller solid black pie that punches the hole.
- **Business logic:** none, it is a charting device.
- **Dependencies:** none.

6.4 Calculation type census for Dashboard 1

Calculation type	Present?	Detail
LOD expressions (<code>FIXED</code> , <code>INCLUDE</code> , <code>EXCLUDE</code>)	No. No <code>{</code> appears in any formula.	
Table calculations	Yes, one type. <code><table-calc ordering-type='Rows' type='PctTotal' /></code>	
Window calculations (<code>WINDOW_SUM</code> , <code>WINDOW_AVG</code> , etc.)	No.	
Running totals	No.	
Percent-of-total	Yes. <code>PctTotal</code> applied to <code>CNT(<dimension>)</code> on all five donut sheets: <code>[pcto:cnt:anaemia:qk]</code> , <code>[pcto:cnt:diabetes:qk]</code> , <code>[pcto:cnt:high_blood_pressure:qk]</code> , <code>[pcto:cnt:sex:qk]</code> , <code>[pcto:cnt:smoking:qk]</code>	
Rank calculations	No.	
Date calculations	No. There is no date field.	
Conditional calculations / IF-THEN-ELSE	No.	
CASE statements	No.	
Parameters used inside calculations	No. There are no parameters in this workbook.	
Aggregations used on shelves	<code>CNT</code> (count), <code>SUM</code> , <code>AVG</code> , and <code>None</code> (continuous dimension)	

Important honesty note. The calculation layer of this workbook is deliberately thin. The heavy lifting is done by aliasing, role conversion, binning and a built-in percent-of-total table calculation, not by written formulas. Do not describe this workbook as containing LODs or complex conditional logic, because it does not.

7. PARAMETERS

There are no parameters in this workbook. ✓ DIRECTLY VERIFIED by absence: there is no `Parameters` data source, no column carrying `param-domain-type`, and no `paramctrl` zone on the dashboard.

Parameter	Data Type	Current Value	Allowed Values	Used By	Purpose
(none)					

8. FILTERS

8.1 Filter inventory

Filter	Field	Type	Values / Condition	Scope	Effect
Sex quick filter	<code>[sex]</code> as <code>[none:sex:ok]</code>	Categorical, <code>function='level-members'</code> , <code>ui-enumeration='all'</code>	All members selected (<code>Female</code> , <code>Male</code>)	<code>filter-group='3'</code> on 14 worksheets , exposed on the dashboard as a dropdown control sourced from <code>Sheet 6 (9)</code>	Filters every donut, every histogram and both scatters simultaneously. Does not touch the five KPI cards.
Total Deaths KPI filter	<code>[DEATH_EVENT]</code> as <code>[none:DEATH_EVENT:ok]</code>	Categorical, <code>function='member'</code> , <code>member='1'</code> , <code>ui-enumeration='inclusive'</code>	Keeps only <code>Death Event = 1</code> (aliased <code>Death</code>)	Worksheet-only: <code>Sheet 15 (2)</code>	Reduces the card's count to deaths only
Total Males KPI filter	<code>[sex]</code> as <code>[none:sex:ok]</code>	Categorical, <code>function='member'</code> , <code>member='1'</code>	Keeps only <code>sex = 1</code> (aliased <code>Male</code>)	Worksheet-only: <code>Sheet 15 (3)</code>	Reduces the card's count to males only
Total Females KPI filter	<code>[sex]</code> as <code>[none:sex:ok]</code>	Categorical, <code>function='member'</code> , <code>member='0'</code>	Keeps only <code>sex = 0</code> (aliased <code>Female</code>)	Worksheet-only: <code>Sheet 15 (4)</code>	Reduces the card's count to females only

8.2 Worksheets affected by the shared Sex filter (`filter-group='3'`)

`Sheet 1`, `Sheet 1 (2)`, `Sheet 1 (3)`, `Sheet 1 (4)`, `Sheet 1 (5)`, `Sheet 6`, `Sheet 6 (2)`, `Sheet 6 (3)`, `Sheet 6 (4)`, `Sheet 6 (5)`, `Sheet 6 (6)`, `Sheet 6 (7)`, `Sheet 6 (8)`, `Sheet 6 (9)`. That is 14 of the 19 worksheets.

Not affected: `Sheet 15`, `Sheet 15 (2)`, `Sheet 15 (3)`, `Sheet 15 (4)`, `Sheet 15 (5)`, the five KPI cards.

Consequence worth knowing (△ inferred): because the KPI cards sit outside `filter-group 3` , selecting `Male` in the Sex dropdown will re-filter all 14 charts but the `Total Individuals` , `Total Deaths` , `Total Males` and `Total Females` cards will not move. That is a real behaviour of the workbook as built, and a likely interview question.

8.3 Filter types NOT present

Filter type	Present?
Data source level filter	No
Extract filter	No
Context filter	No. No filter carries a context flag.
Measure filter	No. Every filter is categorical on a dimension.
Relative date filter	No. There is no date field.
Range / quantitative filter	No
Top N filter	No
Parameter-based filter	No. There are no parameters.
Condition or formula filter	No

9. KPI IDENTIFICATION

Five KPI cards are on the dashboard, each one its own worksheet, all sitting in the top strip at `y=762` (roughly the top 7 percent of the canvas).

KPI 1: Total Individuals

Attribute	Value
Exact displayed name	<code>Total Individuals</code> (the literal string from <code>[Calculation_215609893383794688]</code>)
Tableau field behind the number	<code>[cnt:DEATH_EVENT:qk]</code>
Exact expression	<code>CNT([DEATH_EVENT])</code>
Aggregation	<code>CNT</code> (count of non-null values)
Numerator	Count of rows where <code>DEATH_EVENT</code> is not null
Denominator	Not applicable
Filters affecting it	None. <code>Sheet 15</code> has no filter at all and is outside <code>filter-group 3</code> .
Parameters affecting it	None

Attribute	Value
Time period	Not applicable, the dataset has no dates
Unit	Count of patients
Worksheet	Sheet 15
Dashboard location	Zone id 17, x=421 y=762 w=20892 h=6855 (leftmost KPI, top strip), show-title='false'
Business meaning	Cohort size
Why useful	Every percentage on the dashboard is only interpretable against the base population
What a user learns	How many heart failure patients the entire dashboard is describing
Dependency chain	[Calculation_215609893383794688] supplies the caption text; the number is a direct CNT of a source column. No calculated field feeds the number.

Technical note: counting `DEATH_EVENT` rather than using `COUNT(*)` or a `Number of Records` field is a design choice. It counts non-null `DEATH_EVENT` values, so it equals the cohort size only if `DEATH_EVENT` is never null. The metadata record for `DEATH_EVENT` says `<contains-null>true</contains-null>`, which is Tableau's default optimistic flag for a scanned CSV rather than a measured fact, so this cannot be resolved from the workbook alone. A `COUNT(*)` or an explicit `Number of Records` field would have been immune to that.

KPI 2: Total Deaths

Attribute	Value
Exact displayed name	Total Deaths
Field behind the number	[cnt:DEATH_EVENT:qk]
Exact expression	CNT([DEATH_EVENT]) filtered to DEATH_EVENT = 1
Aggregation	CNT
Numerator	Rows where Death Event = Death
Denominator	Not applicable. No rate is computed. The dashboard shows the count, not the mortality rate.
Filters affecting it	Worksheet filter [none:DEATH_EVENT:ok], function='member', member='1'
Parameters affecting it	None
Unit	Count of patients
Worksheet	Sheet 15 (2)
Dashboard location	Zone id 18, x=21313 y=762 w=19736 h=6855
Business meaning	Absolute mortality in the cohort during follow-up
Why useful	The headline outcome count
What a user learns	How many patients in the cohort died
Dependency chain	Caption from [Calculation_215609893384032257]; number is CNT on a filtered source column

KPI 3: Total Males

Attribute	Value
Exact displayed name	Total Males
Field behind the number	[cnt:sex:qk]
Exact expression	CNT([sex]) filtered to sex = 1
Aggregation	CNT
Filters affecting it	Worksheet filter [none:sex:ok], function='member', member='1'
Unit	Count of patients
Worksheet	Sheet 15 (3)
Dashboard location	Zone id 19, x=41049 y=762 w=18783 h=6855
Business meaning	Male cohort size
What a user learns	The sex composition of the cohort, read together with KPI 4

KPI 4: Total Females

Attribute	Value
Exact displayed name	Total Females
Field behind the number	[cnt:sex:qk]
Exact expression	CNT([sex]) filtered to sex = 0
Aggregation	CNT
Filters affecting it	Worksheet filter [none:sex:ok], function='member', member='0'
Unit	Count of patients
Worksheet	Sheet 15 (4)
Dashboard location	Zone id 20, x=59832 y=762 w=20217 h=6855
Business meaning	Female cohort size
What a user learns	Whether the cohort is sex-balanced, which conditions how the Sex donut should be read

KPI 5: Average Age

Attribute	Value
Exact displayed name	Average Age
Field behind the number	[avg:age:qk]
Exact expression	AVG([age])

Attribute	Value
Aggregation	AVG
Filters affecting it	None. Sheet 15 (5) has no filter and is outside filter-group 3.
Unit	Years
Worksheet	Sheet 15 (5)
Dashboard location	Zone id 21 , x=80049 y=762 w=19530 h=6855 (rightmost KPI)
Business meaning	Central tendency of cohort age
What a user learns	Roughly how old this heart failure population is, which frames the Age histogram below it

KPIs that are NOT present

There is **no mortality rate**, **no survival rate**, **no percentage KPI**, and **no target or benchmark comparison** on this dashboard. The percent-of-total figures exist only inside the donut labels, not as KPI cards. ✓ VERIFIED by absence.

10. WORKSHEET / SHEET-BY-SHEET ANALYSIS

19 worksheets exist and **all 19 are placed on Dashboard 1**. None is orphaned.

Global formatting shared by the chart worksheets (Sheet 1 family and Sheet 6 family), read from each <style> block:

Style element	Setting
table background-color	#000000
worksheet font	bold, #ffffff, size 8
zeroline	stroke-size 0, line-visibility off
table-div (grid lines), cols scope	stroke-size 0, line-visibility off
title	1px solid #ffffff border
header subtotal border	1px solid #ffffff
cell size on donut sheets	width 290 , height 223

10.1 to 10.5: The five donut charts (Sheet 1 family)

All five are structurally identical. Only the comorbidity dimension changes.

Worksheet	Title (exact)	Dimension used	Angle field	Percent-of-total field
Sheet 1	Anaemia-Survival Stats	[anaemia]	[cnt:anaemia:qk]	[pcto:cnt:anaemia:qk]
Sheet 1 (2)	Diabetes-Survival Stats	[diabetes]	[cnt:diabetes:qk]	[pcto:cnt:diabetes:qk]
Sheet 1 (3)	High Blood Pressure-Survival Stats	[high_blood_pressure]	[cnt:high_blood_pressure:qk]	[pcto:cnt:high_blood_pressure:qk]
Sheet 1 (4)	Sex-Survival Stats	[sex]	[cnt:sex:qk]	[pcto:cnt:sex:qk]
Sheet 1 (5)	Smoking-Survival Stats	[smoking]	[cnt:smoking:qk]	[pcto:cnt:smoking:qk]

Worksheet: Sheet 1 (representative of all five)

Property	Value
Purpose	Show how patients split across Death Event and Anaemia at the same time, with both a count and a share
Visualization type	Donut chart , built as a dual-axis pie
Data source	heart_failure_clinical_records_dataset
Rows shelf	(SUM([0]) + SUM([0])), that is the 0 calc placed twice to create a dual axis
Columns shelf	empty
Marks type	Pie on all three panes
Marks > Color	Pane 1: [none:DEATH_EVENT:ok] and [none:anaemia:ok] (two fields on Color, producing a 4-way crossed palette). Pane 2: solid #000000 via mark-color format, no field.
Marks > Size	Pane 1 size = 0.69165748357772827. Pane 2 size = 0.42779004573822021. mark-sizing-setting='marks-scaling-off' on both. The smaller black pie on the second axis is what creates the donut hole.
Marks > Angle (wedge-size)	[cnt:anaemia:qk], that is CNT([anaemia])
Marks > Label (text)	Three fields: [none:anaemia:ok], [none:DEATH_EVENT:ok], [pcto:cnt:anaemia:qk]. Labels shown (mark-labels-show = true) and culled (mark-labels-cull = true) on pane 1; hidden on pane 2.
Marks > Detail	none beyond the above
Marks > Tooltip	No custom tooltip. Tableau default applies.
Filters	[none:sex:ok], level-members, all, filter-group='3'
Pages	none
Parameters	none
Sets / Groups	none
Calculated fields used	[Calculation_144326355385667585] (the 0 axis placeholder)
Table calculations	PctTotal, ordering-type='Rows', applied to CNT([anaemia])
Sorting	none defined
Reference lines / trend lines / forecasting	none
Analytics features	none

Property	Value
Axis configuration	Both row axes: <code>major-show='false'</code> , <code>major-spacing='2147483647'</code> , axis <code>title=''</code> (blank), axis <code>width=48</code> . Axes are effectively hidden.
Number formatting	None specified. No <code>number-format</code> attribute exists anywhere in this workbook, so defaults apply.
Date formatting	Not applicable
Dual axis	Yes. Two <code>SUM([0])</code> fields on Rows, panes with <code>y-index='1'</code> on the second.
Synchronized axis	Both axes are the same constant <code>SUM([0])</code> expression, so they coincide by construction. There is no separate <code><synchronize></code> flag in the XML. $\triangle$ INFERRED
Measure Names / Measure Values	Not used
Color palette	<code>palette='summer_10_0'</code> with an explicit override map

Exact color map on the donut sheets (identical on all five):

Death Event bucket	Comorbidity bucket	Hex
0 (Survive)	0	#8fb202 (olive green)
0 (Survive)	1	#b9ca5d (light green)
1 (Death)	0	#cf3e53 (deep red)
1 (Death)	1	#f1788d (pink)

Plain English, how the donut works. A constant measure of zero is placed on Rows twice, which gives Tableau two identical vertical axes to draw on. Both mark types are set to Pie. The first pie is sized `0.69` and is coloured by the crossed combination of outcome and comorbidity, so it has four wedges: survived without the condition, survived with it, died without it, died with it. Wedge angle is the count of patients in each combination, and each wedge is labelled with the comorbidity value, the outcome value and the wedge's percent of the total count computed across rows. The second pie is sized `0.43` , coloured solid black to match the dashboard background, and has its labels turned off. Overlaying the small black pie on the large coloured one punches a hole in the middle, which is how a donut is made in Tableau without a dedicated donut mark type.

10.6 to 10.12: The seven histograms (Sheet 6 family, part 1)

All seven are structurally identical: a bin on Columns, a count on Rows, outcome on Color.

Worksheet	Title (exact)	Columns shelf	Rows shelf
Sheet 6	Age - Survival Stats	[none:Age (bin):qk]	[cnt:age:qk]
Sheet 6 (2)	Creatinine Phosphokinase - Survival Stats	[none:Creatinine Phosphokinase (bin):qk]	[cnt:creatinine_phosphokinase:qk]
Sheet 6 (3)	Ejection Fraction - Survival Stats	[none:Ejection Fraction (bin):qk]	[cnt:ejection_fraction:qk]
Sheet 6 (4)	Platelets - Survival Stats	[none:Platelets (bin):qk]	[cnt:platelets:qk]

Worksheet	Title (exact)	Columns shelf	Rows shelf
Sheet 6 (5)	Serum Creatinine - Survival Stats	[none:Serum Creatinine (bin):qk]	[cnt:serum_creatinine:qk]
Sheet 6 (6)	Serum Sodium - Survival Stats	[none:Serum Sodium (bin):qk]	[cnt:serum_sodium:qk]
Sheet 6 (7)	Time - Survival Stats	[none:Time (bin):qk]	[cnt:time:qk]

Worksheet: Sheet 6 (representative of all seven)

Property	Value
Purpose	Show the distribution of age across the cohort, split by survival outcome
Visualization type	Stacked histogram (bar marks over a bin dimension)
Marks type	<code>Automatic</code> . With a continuous measure on Rows and a discrete bin on Columns, Tableau resolves Automatic to Bar . $\triangle$ INFERRED from Tableau's mark resolution rules; the XML stores only <code>Automatic</code> .
Marks > Color	[none:DEATH_EVENT:ok]
Color map (all seven sheets)	0 (Survive) #59a14f (green), 1 (Death) #e15759 (red). <code>type='palette'</code> , no named palette override.
Marks > Size / Label / Detail	none
Marks > Tooltip	No custom tooltip, Tableau default
Filters	[none:sex:ok], level-members, all, <code>filter-group='3'</code>
Sorting	none
Reference lines	none
Analytics features	none
Dual axis	no
Measure Names / Values	not used
Aggregate Measures	<code>true</code>
Color legend	Only Sheet 6 exposes its color legend on the dashboard (zone id 25). The other six histograms reuse the same color scheme without their own legend.

Plain English. Age is bucketed into 4.48-year bins on the horizontal axis. The bar height is the count of patients in that bin. Each bar is split into a green segment (survived) and a red segment (died), stacked. Reading left to right shows where in the age range the red proportion grows. The same construction is repeated for creatinine phosphokinase, ejection fraction, platelets, serum creatinine, serum sodium and follow-up time, which means seven biomarkers are profiled in exactly the same visual grammar.

10.13: Sheet 6 (8) Age vs Time scatter

Property	Value
Title (exact)	Age-Time - Survival Stats
Purpose	Relate patient age to follow-up time, coloured by outcome
Visualization type	Scatter plot
Columns shelf	[none:age:qk] , that is Age used with no aggregation (continuous dimension)
Rows shelf	[none:time:qk] , that is Time used with no aggregation
Marks type	Automatic
Marks > Color	[none:DEATH_EVENT:ok] , green #59a14f / red #e15759
Marks > Size / Label / Detail	none
Tooltip	default
Filters	[none:sex:ok] , level-members, all, filter-group='3'
Aggregate Measures	true
Reference lines / trend lines / forecast	none
Dual axis	no
Dashboard zone	id 32 , x=41049 y=66285 w=39002 h=32953. This is the widest chart zone on the dashboard , twice the width of a normal tile.

Plain English. Both age and time are placed on the shelves as continuous *dimensions* rather than aggregated measures, so each distinct age-and-follow-up-time combination draws its own mark instead of being rolled up. Colour shows whether that combination corresponds to survivors or deaths. The wide zone gives the horizontal age axis room to spread out.

10.14: Sheet 6 (9) Age by outcome and sex

Property	Value
Title (exact)	Age-Sex - Survival Stats
Purpose	Compare the age distribution across the four outcome-by-sex cells
Visualization type	Disaggregated dot plot with an average reference line per cell
Columns shelf	([none:DEATH_EVENT:ok] / [none:sex:ok]) , a nested discrete split: outcome first, then sex inside it, giving 4 cells
Rows shelf	[avg:age:qk] , that is AVG([age])
Marks type	Circle (explicitly set, not Automatic)
Aggregate Measures	false. This is the only worksheet in the workbook where Analysis > Aggregate Measures is switched off, so each underlying record is plotted as its own circle.
Marks > Color	none defined on this sheet

Property	Value
Marks > Size / Label / Detail	none
Reference line	Yes. <code><reference-line axis-column='[avg:age:qk]' formula='average' value-column='[avg:age:qk]' scope='per-cell' label-type='automatic' id='refline0' z-order='1' probability='95' symmetric='false' boxplot-whisker-type='standard' boxplot-mark-exclusion='false' enable-instant-analytics='true' /></code>
Reference line meaning	An average line drawn per cell , so each of the four outcome-by-sex cells gets its own mean age line
Filters	<code>[none:sex:ok]</code> , level-members, all, <code>filter-group='3'</code> . This worksheet is the source of the dashboard's Sex dropdown control.
Field labels	<code>display-field-labels</code> set to <code>false</code> for cols scope, so the column header captions are suppressed
Dual axis	no

Plain English. The horizontal axis is split into four groups: survived-female, survived-male, died-female, died-male (nesting outcome over sex). Because aggregation is turned off, every patient appears as an individual circle at their own age. On top of each group a horizontal line marks that group's average age. This is the one view in the dashboard that shows within-group spread rather than just counts, and it is also the view that hosts the global Sex filter control.

10.15 to 10.19: The five KPI worksheets (**Sheet 15** family)

Worksheet	Text shelf field 1 (caption)	Text shelf field 2 (value)	Worksheet filter
Sheet 15	<code>[none:Calculation_215609893383794688:nk] = "Total Individuals"</code>	<code>[cnt:DEATH_EVENT:qk]</code>	none
Sheet 15 (2)	<code>[none:Calculation_215609893384032257:nk] = "Total Deaths"</code>	<code>[cnt:DEATH_EVENT:qk]</code>	DEATH_EVENT member 1
Sheet 15 (3)	<code>[none:Calculation_215609893384187906:nk] = "Total Males"</code>	<code>[cnt:sex:qk]</code>	sex member 1
Sheet 15 (4)	<code>[none:Calculation_215609893384642563:nk] = "Total Females"</code>	<code>[cnt:sex:qk]</code>	sex member 0
Sheet 15 (5)	<code>[none:Calculation_215609893384794116:nk] = "Average Age"</code>	<code>[avg:age:qk]</code>	none

Shared properties for all five:

Property	Value
Visualization type	Text / BAN card (big aggregate number)
Marks type	<code>Automatic</code> (resolves to Text, since only Text encodings are present)
Rows shelf	empty
Columns shelf	empty
Marks > Text	two fields: the constant caption string, then the aggregate
Mark labels	<code>mark-labels-show = true, mark-labels-cull = true</code>
Marks > Color / Size / Detail	none
Tooltip	default

Property	Value
Title on dashboard	suppressed, all five zones carry <code>show-title='false'</code>
Aggregate Measures	<code>true</code>
Number formatting	none specified

Plain English. Each card is an empty viz with two things on the Text shelf: a string constant that acts as the label and a single aggregate that acts as the number. Because rows and columns are empty, Tableau draws exactly one mark, which renders as a centred block of text. Titles are hidden on the dashboard so the constant string is the only caption visible.

11. VISUALIZATION-BY-VISUALIZATION BREAKDOWN

#	Visual	What it shows	Fields driving it	Dimensions compared	Measure analysed	Aggregation	Filters affecting it	Calculations affecting it	Business question answered	Why appropriate
1	KPI: Total Individuals	Cohort size	<code>DEATH_EVENT</code>	none	count of patients	<code>CNT</code>	none	none	How big is the population?	A single number is the right form for a scalar
2	KPI: Total Deaths	Deaths in cohort	<code>DEATH_EVENT</code>	none	count of patients	<code>CNT</code>	<code>DEATH_EVENT = 1</code>	none	How many died?	Same
3	KPI: Total Males	Male count	<code>sex</code>	none	count of patients	<code>CNT</code>	<code>sex = 1</code>	none	Sex composition	Same
4	KPI: Total Females	Female count	<code>sex</code>	none	count of patients	<code>CNT</code>	<code>sex = 0</code>	none	Sex composition	Same
5	KPI: Average Age	Mean age	<code>age</code>	none	age	<code>AVG</code>	none	none	How old is this population?	Same
6	Donut: Anaemia-Survival Stats	4-way split of the cohort across outcome and anaemia status	<code>DEATH_EVENT</code> , <code>anaemia</code>	Outcome x Anaemia	count, plus share of total	<code>CNT + PctTotal</code>	Sex filter	<code>[0]</code> axis calc, <code>PctTotal</code>	Do anaemic patients die more often?	A donut with a crossed colour encoding shows composition and share in one mark; the hole leaves room for the dark theme
7	Donut: Diabetes-Survival Stats	Same for diabetes	<code>DEATH_EVENT</code> , <code>diabetes</code>	Outcome x Diabetes	count + share	<code>CNT + PctTotal</code>	Sex filter	same	Does diabetes associate with mortality?	Same
8	Donut: High Blood Pressure-Survival Stats	Same for hypertension	<code>DEATH_EVENT</code> , <code>high_blood_pressure</code>	Outcome x HBP	count + share	<code>CNT + PctTotal</code>	Sex filter	same	Does hypertension associate with mortality?	Same
9	Donut: Sex-Survival Stats	Same for sex	<code>DEATH_EVENT</code> , <code>sex</code>	Outcome x Sex	count + share	<code>CNT + PctTotal</code>	Sex filter	same	Does outcome differ by sex?	Same

#	Visual	What it shows	Fields driving it	Dimensions compared	Measure analysed	Aggregation	Filters affecting it	Calculations affecting it	Business question answered	Why appropriate
10	Donut: Smoking-Survival Stats	Same for smoking	DEATH_EVENT, smoking	Outcome x Smoking	count + share	CNT + PctTotal	Sex filter	same	Does smoking associate with mortality?	Same
11	Histogram: Age - Survival Stats	Age distribution split by outcome	Age (bin), DEATH_EVENT	Age buckets	count of patients	CNT	Sex filter	Age (bin)	Where in the age range do deaths cluster?	A stacked histogram is the standard form for a binned continuous variable split by a binary class
12	Histogram: Creatinine Phosphokinase	CPK distribution by outcome	Creatinine Phosphokinase (bin), DEATH_EVENT	CPK buckets	count	CNT	Sex filter	bin	Is CPK elevated among those who died?	Same
13	Histogram: Ejection Fraction	EF distribution by outcome	Ejection Fraction (bin), DEATH_EVENT	EF buckets	count	CNT	Sex filter	bin	Is low ejection fraction associated with death?	Same
14	Histogram: Platelets	Platelet distribution by outcome	Platelets (bin), DEATH_EVENT	Platelet buckets	count	CNT	Sex filter	bin	Does platelet count separate the outcomes?	Same
15	Histogram: Serum Creatinine	Creatinine distribution by outcome	Serum Creatinine (bin), DEATH_EVENT	Creatinine buckets	count	CNT	Sex filter	bin	Does renal function associate with mortality?	Same
16	Histogram: Serum Sodium	Sodium distribution by outcome	Serum Sodium (bin), DEATH_EVENT	Sodium buckets	count	CNT	Sex filter	bin	Does hyponatraemia associate with mortality?	Same
17	Histogram: Time	Follow-up time distribution by outcome	Time (bin), DEATH_EVENT	Time buckets	count	CNT	Sex filter	bin	How does follow-up duration relate to outcome?	Same
18	Scatter: Age-Time - Survival Stats	Age against follow-up time	age, time, DEATH_EVENT	none (two continuous axes)	age, time	none (unaggregated)	Sex filter	none	Do deaths cluster at particular age-and-follow-up combinations?	A scatter is the right form for two continuous variables with a categorical colour
19	Dot plot: Age-Sex - Survival Stats	Individual ages in four outcome-by-sex cells with per-cell mean lines	age, DEATH_EVENT, sex	Outcome nested over Sex	age	disaggregated, plus an average reference line	Sex filter	reference line formula='average', scope='per-cell'	Does mean age differ across outcome and sex?	Showing individual points plus a mean line conveys both central tendency and spread, which a bar of averages would hide

12. DASHBOARD LAYOUT

Property	Value
Dashboard name	Dashboard 1
Size	1900 x 1050 px, <code>sizing-mode='range'</code> , min and max both 1900 x 1050
Background	<code>#000000</code>
Layout style	Fully tiled. There are no floating objects. The whole dashboard is built from nested flow containers.
Container structure	<code>layout-basic</code> (id 4) > <code>layout-flow param='horz'</code> (id 35) > <code>layout-flow param='vert'</code> (id 16) > <code>layout-flow param='horz'</code> (id 7) > <code>layout-basic</code> (id 5) which holds all 19 worksheet zones
Outer container padding	<code>margin='8'</code>
Worksheet zone padding	<code>margin='4'</code> on every worksheet zone
Borders	<code>border-style='none'</code> , <code>border-width='0'</code> , <code>border-color='#000000'</code> on every zone
Device layouts	One auto-generated Phone layout, <code>sizing-mode='vscroll'</code> , height 4450 px, <code>auto-generated='true'</code> . It re-stacks the same 21 objects into a single vertical flow container (id 41). No Tablet layout.

Object census

Object type	Count	Detail
Worksheets	19	All 19 worksheets in the workbook
Color legend	1	Zone id <code>25</code> , <code>type-v2='color'</code> , <code>param='[none:DEATH_EVENT:ok]'</code> , sourced from <code>Sheet 6</code> , <code>show-title='false'</code> , at x=74105 y=36667 w=5474 h=3905
Filter control	1	Zone id <code>36</code> , <code>type-v2='filter'</code> , <code>mode='dropdown'</code> , <code>param='[none:sex:ok]'</code> , sourced from <code>Sheet 6 (9)</code> , at x=34842 y=37143 w=5474 h=4762
Text objects	0	There is no title text object and no annotation
Images	0	
Blank objects	0	
Navigation buttons	0	
Extension objects	0	
Web page objects	0	
Parameter controls	0	No parameters exist

Reconstructed hierarchy

```
Dashboard 1 (1900 x 1050, black, tiled)
├─ layout-basic [id 4, margin 8]
│   └─ layout-flow horz [id 35]
│       └─ layout-flow vert [id 16]
│           └─ layout-flow horz [id 7]
│               └─ layout-basic [id 5]
│                   └─ ROW 1 y=762, h=6855 KPI STRIP (titles hidden)
│                       └─ Sheet 15 Total Individuals x=421 w=20892
│                       └─ Sheet 15 (2) Total Deaths x=21313 w=19736
│                       └─ Sheet 15 (3) Total Males x=41049 w=18783
│                       └─ Sheet 15 (4) Total Females x=59832 w=20217
│                       └─ Sheet 15 (5) Average Age x=80049 w=19530
│                   └─ ROW 2 y=7617, h=25240 DONUT STRIP (comorbidity vs outcome)
│                       └─ Sheet 1 Anaemia x=421 w=20892
│                       └─ Sheet 1 (3) High Blood Pressure x=21313 w=19736
│                       └─ Sheet 1 (5) Smoking x=41049 w=18792
│                       └─ Sheet 1 (2) Diabetes x=59841 w=20208
│                       └─ Sheet 1 (4) Sex x=80049 w=19530
│                   └─ ROW 3 y=32857, h~33428 DISTRIBUTION STRIP A
│                       └─ Sheet 6 Age histogram x=421 w=20892
│                           └─ [floating-in-flow] color legend "Death Event" x=74105 y=36667
│                       └─ Sheet 6 (2) CPK histogram x=21313 w=19736
│                       └─ Sheet 6 (9) Age-Sex dot plot x=41049 w=18792
│                           └─ [control] Sex filter dropdown x=34842 y=37143
│                       └─ Sheet 6 (3) Ejection Fraction x=59841 w=20210
│                       └─ Sheet 6 (4) Platelets x=80051 w=19528
│                   └─ ROW 4 y~66285, h~32953 DISTRIBUTION STRIP B
│                       └─ Sheet 6 (6) Serum Sodium x=421 w=20892
│                       └─ Sheet 6 (7) Time histogram x=21313 w=19736
│                       └─ Sheet 6 (8) Age-Time scatter x=41049 w=39002 <-- double width
│                       └─ Sheet 6 (5) Serum Creatinine x=80051 w=19528
```

Note on reading the numbers: zone **x**, **y**, **w**, **h** are stored in hundred-thousandths of the dashboard dimension. Multiply by 1900/100000 for pixels horizontally and 1050/100000 vertically. For example **Sheet 15** at **x=421 w=20892** is roughly 8 px from the left and 397 px wide.

Headers and footers: there are none as distinct objects. The KPI strip functions as a header visually, but it is five worksheets, not a header object. ✓ VERIFIED.

13. DASHBOARD ACTIONS / INTERACTIVITY

There is exactly **one** action in the workbook.

Attribute	Value
Caption	Highlight 1 (generated)
Internal name	[Action1_F214FEC6D9404714A6A883192F2B2305]
Action type	Highlight (command='tsc:brush')
Trigger	<activation type='on-select' auto-clear='true' />, that is, select a mark, and deselecting clears it
Source	<source dashboard='Dashboard 1' type='sheet' />, so the source is the whole dashboard, every sheet on it
Target	param name='target' value='Dashboard 1', the whole dashboard
Fields used	param name='field-captions' value='Death Event'
Result / effect	Selecting any mark highlights, across all 19 worksheets at once, every other mark that shares the same Death Event value. Selecting a red wedge on the Anaemia donut dims every non- Death mark on all the histograms, both scatters and the other four donuts.
Caption suffix (generated)	Indicates Tableau auto-named it when the user created a highlight action on a field. △ INFERRED

Interactivity types NOT present

Interactivity	Present?
Filter actions	No
URL actions	No
Parameter actions	No (no parameters exist)
Set actions	No (no sets exist)
Navigation / Go to Sheet actions	No
Sheet swapping	No
Dashboard-to-dashboard navigation	No (there is only one dashboard)
Drill-down hierarchies	No (no hierarchies defined)
Viz-in-tooltip	No
Custom tooltip interactions	No

Interactive controls available to the end user: the Sex dropdown filter (affecting 14 sheets), the Death Event highlight action (affecting all 19), and the Death Event color legend (which in Tableau also supports highlight-on-click).

14. TOOLTIPS

No worksheet in this workbook has a customised tooltip. There is not a single `<customized-tooltip>` element in the file. ✓ DIRECTLY VERIFIED by absence.

Worksheet	Custom tooltip?	Calculated values in tooltip?	Custom descriptions?	Viz-in-tooltip?	Conditional behaviour?
All 19 worksheets	No	No	No	No	No

What users will actually see. Tableau's default tooltip lists whatever is on the shelves for that mark. So for example:

- On a donut wedge: `Anaemia` , `Death Event` , `CNT(Anaemia)` , `% of Total CNT(Anaemia)` .
- On a histogram bar: `Age (bin)` , `Death Event` , `CNT(Age)` .
- On the Age-Sex dot plot: `Death Event` , `Sex` , `Age` .

These are $\triangle$ INFERRED from the encodings, since the tooltip content itself is not stored when it is left at default.

Worksheet titles that ARE customised (these are stored, unlike tooltips). All use bold white 11pt text with `fontalignment='1'` (centred):

Worksheet	Exact title text
Sheet 1	Anaemia-Survival Stats
Sheet 1 (2)	Diabetes-Survival Stats
Sheet 1 (3)	High Blood Pressure-Survival Stats
Sheet 1 (4)	Sex-Survival Stats
Sheet 1 (5)	Smoking-Survival Stats
Sheet 6	Age - Survival Stats
Sheet 6 (2)	Creatinine Phosphokinase - Survival Stats
Sheet 6 (3)	Ejection Fraction - Survival Stats
Sheet 6 (4)	Platelets - Survival Stats
Sheet 6 (5)	Serum Creatinine - Survival Stats
Sheet 6 (6)	Serum Sodium - Survival Stats
Sheet 6 (7)	Time - Survival Stats
Sheet 6 (8)	Age-Time - Survival Stats
Sheet 6 (9)	Age-Sex - Survival Stats
The five Sheet 15 cards	No custom title; titles are hidden on the dashboard

15. BUSINESS LOGIC

Stated carefully: the workbook records *what was built*, not *why*. Everything below is either the mechanical logic that is in the file (✓) or a reading of it (△).

What business problem was identified (△ INFERRED). Heart failure patients vary widely in outcome, and clinicians and analysts want to know which recorded attributes travel with mortality. The dashboard treats that as a profiling problem rather than a prediction problem: it does not model risk, it displays how the cohort splits on every available attribute.

What data was needed (✓). A single patient-level clinical record file with one row per patient, containing a binary mortality outcome (`DEATH_EVENT`), a follow-up period (`time`), four binary comorbidity or lifestyle flags (`anaemia` , `diabetes` , `high_blood_pressure` , `smoking`), a demographic flag (`sex`), and six continuous clinical measurements (`age` , `creatinine_phosphokinase` , `ejection_fraction` , `platelets` , `serum_creatinine` , `serum_sodium`).

How the data was transformed (✓). 1. The CSV was read with a `textscan` connection using UTF-8, comma separator, header row, `en_US` locale. 2. The whole table was extracted to Hyper with no row limit and no extract filter. 3. All 13 columns were given display captions. 4. Six binary integer columns were converted from measure to dimension with ordinal type. 5. Value aliases decoded `0/1` into clinical language on all six. 6. Seven bins were created on the continuous measures using Tableau's suggested bin sizes. 7. Five string-literal calculated fields were created to caption the KPI cards. 8. One constant `0` measure was created to enable dual-axis donuts.

What KPIs were selected (✓). Cohort size, death count, male count, female count, mean age. All are counts or a mean. No rates, no ratios, no targets.

What dimensions were analysed (✓). `Death Event` (the outcome, present on every chart), `Sex` , `Anaemia` , `Diabetes` , `High Blood Pressure` , `Smoking` , plus seven binned continuous dimensions.

What comparisons were made (✓). - Outcome crossed with each binary attribute, as a 4-way donut with counts and percent of total (5 charts). - Outcome across the distribution of each continuous biomarker, as a stacked histogram (7 charts). - Outcome against two continuous variables at once, as a scatter (1 chart). - Mean age compared across four outcome-by-sex cells with individual points visible (1 chart).

What trends were intended to be identified (△ INFERRED). The design points at three things: whether the red (`Death`) proportion is visually larger in the presence of each comorbidity; where along each biomarker's range the red mass concentrates; and whether outcome separation is different for men and women. Note that `time` is a follow-up duration, not a calendar date, so **this dashboard contains no time series and cannot show a trend over time**. That is a distinction worth stating plainly rather than glossing.

What decisions a business user could make (△ INFERRED). Identify which recorded attributes deserve a formal statistical test or a predictive model; decide which biomarker thresholds are worth investigating for a risk score; decide whether a sex-stratified analysis is warranted; and communicate cohort composition to a clinical stakeholder.

16. KEY INSIGHTS THE DASHBOARD IS DESIGNED TO ENABLE

No actual data values were read, so no findings are claimed.

A. Insights explicitly encoded by the dashboard

These are things the dashboard computes and displays as a matter of construction:

1. The cohort size, death count, male count, female count and mean age are computed and displayed as five standing figures.
2. For each of anaemia, diabetes, high blood pressure, sex and smoking, the dashboard computes the count and the **percent of total** for each of the four combinations of that attribute with the survival outcome.
3. For each of age, creatinine phosphokinase, ejection fraction, platelets, serum creatinine, serum sodium and follow-up time, the dashboard computes the count of patients per fixed-width bin, stacked by outcome.
4. For the four outcome-by-sex cells, the dashboard computes and draws the **mean age per cell** as a reference line.
5. Selecting any mark propagates a highlight on **Death Event** across all 19 views at once.

B. Questions the dashboard enables users to answer

- "How large is this heart failure cohort, and how is it split by sex?"
- "How many patients died during follow-up?"
- "What share of anaemic patients died, compared with the share of non-anaemic patients who died?" (and the same for diabetes, hypertension, smoking and sex)
- "Across the range of ejection fraction, where do deaths concentrate?" (and the same for the other six binned measures)
- "Is the age profile of patients who died different from those who survived, and does that differ by sex?"
- "If I restrict to men only, does every one of those patterns change?" (via the Sex dropdown, noting the KPI cards will not respond)
- "Which marks anywhere on the dashboard belong to the same outcome class as the one I just clicked?" (via the highlight action)

C. Actual findings verifiable from the available data

None can be stated. The underlying values live only in the binary `.hyper` extract, and no Hyper reader is available in this environment. No mortality rate, no distribution shape and no correlation can be verified from the files supplied. X NOT AVAILABLE.

17. TECHNICAL IMPLEMENTATION SUMMARY

DATA SOURCE

heart_failure_clinical_records_dataset.csv
13 columns, one row per patient
/Users/atharvadevne/UIIC/Business Data Visualization/Heart Fail Prediction

|

v

DATA CONNECTION

Tableau textscan (flat file) connector
UTF-8, comma-separated, header row, en_US locale
wrapped in federated.15n4u4511e3xfk1749m9rli7l4e1
Extract enabled, count = -1 (all records), no extract filter
Materialised to #TableauTemp_1yyvr7n0uz5nmh12wm62y1t1ntc5.hyper

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DATA MODEL

Single table. No joins, no relationships, no unions, no blends.

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DATA CLEANING (all inside Tableau, no external prep)

13 field captions applied
6 integer flags converted measure -> dimension (ordinal)
6 fields value-aliased: 0/1 -> Survive/Death, Negative/Positive,
Female/Male, " Non-Smoker"/Smoker
7 bins created on the continuous measures (Tableau-suggested sizes)

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CALCULATED FIELDS (13 total, no LODs, no IF/CASE)

7 x bin calculations
5 x string literal captions for KPI cards
1 x constant 0 measure to enable dual-axis donuts
+ 1 built-in table calculation: Percent of Total (PctTotal, along Rows)

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KPIs (5)

CNT(DEATH_EVENT) -> Total Individuals
CNT(DEATH_EVENT) filtered DEATH_EVENT=1 -> Total Deaths
CNT(sex) filtered sex=1 -> Total Males
CNT(sex) filtered sex=0 -> Total Females
AVG(age) -> Average Age

|

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WORKSHEETS (19)

5 dual-axis donuts (Pie + Pie, crossed colour, PctTotal labels)
7 stacked histograms (bin on Columns, CNT on Rows, outcome on Colour)
1 scatter (age vs time, unaggregated, outcome on Colour)
1 disaggregated dot plot with per-cell average reference line
5 text/BAN KPI cards

```
|
v
DASHBOARD (1)
  "Dashboard 1", 1900 x 1050 range sizing, black background
  Fully tiled, nested horizontal/vertical flow containers
  19 worksheet zones + 1 color legend + 1 filter dropdown
  Auto-generated Phone layout (vscroll, 4450 px)
  |
  v
  FILTERS / ACTIONS
    Sex dropdown -> filter-group 3 -> 14 worksheets (NOT the KPI cards)
    3 hard-coded worksheet filters on the Deaths/Males/Females cards
    1 highlight action on "Death Event", dashboard-wide, on-select, auto-clear
    |
    v
    FINAL USER EXPERIENCE
      A single dark screen showing cohort headline numbers, five comorbidity
      donuts, seven biomarker histograms and two scatter-style views, all colour
      coded green for survival and red for death, with one dropdown to restrict
      by sex and click-to-highlight linking every view on the outcome field.
```

18. INTERVIEW EXPLANATION

Use this as a spoken narrative. Everything in it is supported by the workbook.

"This one is a heart failure survival dashboard I built for myself, to get properly fluent in Tableau's semantic layer and in building chart types that Tableau does not give you out of the box. The dataset is a public clinical records set from Kaggle, 13 columns, one row per patient: a binary mortality flag, a follow-up period, four comorbidity flags for anaemia, diabetes, high blood pressure and smoking, plus sex, age and five clinical measurements like ejection fraction, serum creatinine and serum sodium.

I built it because the raw file is basically unreadable as a table. It is all zeros and ones and unlabelled biomarker columns. The question I wanted the dashboard to answer was simple to state and hard to see in the data: for every attribute we record about a heart failure patient, how does the population split between survival and death?

The data came in as a single CSV read through Tableau's flat file connector, comma-separated with a header row, and I extracted the whole thing to Hyper with no row limit so the workbook is self-contained and fast. There is only one table, so there is no join or relationship layer at all, and I would not pretend otherwise. That was a deliberate scope decision. The complexity in this project is not in the data model, it is in the preparation and the visual grammar.

The preparation is where most of the real work sits. First I renamed all 13 columns to readable captions. Then, and this is the important one, Tableau imports those zero-one flag columns as measures, so it wants to sum them, which is meaningless. I converted six of them to ordinal dimensions so they could drive colour, labels and filters properly. Then I aliased every one of those binary values at the data source level, so zero and one became Survive and Death, Negative and Positive, Female and Male, Non-Smoker and Smoker. Because I did that at the source rather than in each sheet, all 19 worksheets inherit it and I only had to define it once. Finally I created seven bins on the continuous measures so I could draw distributions.

For KPIs I picked cohort size, total deaths, male count, female count and average age. I built each one as its own worksheet with a string constant on the Text shelf for the caption and a single aggregate next to it, then hid the titles on the dashboard. Total Deaths, Total Males and Total Females each carry a hard worksheet filter on the member value, so the card is doing the filtering, not the sheet.

The calculations are deliberately light. Seven bins, five string literals for the card captions, and one constant zero measure. That last one is a technique rather than a business calculation: I put the zero measure on Rows twice to create a dual axis, set both mark types to Pie, made the second pie smaller and solid black to match the background, and turned its labels off. That is how you build a donut in Tableau, since there is no donut mark type. For the wedge labels I used the built-in Percent of Total table calculation computed along Rows, so each wedge shows both its count and its share.

For the distributions I used the same grammar seven times: the bin on Columns, a count on Rows, and the death event on Colour, green for survived and red for died. Repeating one visual form across seven biomarkers means a reader learns how to read the chart once and then reads six more for free.

There are two views that break the pattern on purpose. One is a scatter of age against follow-up time, where I put both fields on the shelves as continuous dimensions rather than aggregated measures so every record draws its own mark. The other is a dot plot of age split by outcome and then by sex, where I switched off Aggregate Measures so you see the

individual spread, and added an average reference line scoped per cell so each of the four groups gets its own mean line. That view shows distribution, not just counts.

For the dashboard itself, 1900 by 1050, black background, fully tiled with nested flow containers rather than floating objects, so it reflows cleanly. Nineteen worksheets, one colour legend and one sex filter dropdown, and Tableau's auto-generated phone layout on top.

For interactivity I went with one highlight action on Death Event, scoped to the whole dashboard, triggered on select with auto-clear. Click any mark anywhere and every other view dims everything that is not the same outcome class. Combined with the sex dropdown, which is applied to the 14 chart sheets, that gives you two ways to interrogate the same picture.

If I were to extend it, the honest gaps are these: I would add a mortality rate KPI rather than only counts, I would put the sex filter on the KPI cards too so the whole dashboard moves together, I would set my own bin sizes instead of accepting Tableau's suggested widths, and I would fix two data quality issues I can see in my own aliases, a misspelling of Survive and a leading space on Non-Smoker.

Skills-wise this shows data source level aliasing and role management, binning, dual-axis chart construction, table calculations, reference lines, disaggregated views, filter scoping across sheet groups, dashboard action design, container-based layout, and a consistent dark visual system across nineteen views."

19. RESUME-READY PROJECT DESCRIPTION

Only verified facts. No invented percentages, record counts or outcomes.

Healthcare Analytics Dashboard, Heart Failure Survival Profiling (Tableau)

- Built a 19-worksheet Tableau dashboard profiling heart failure patient survival across 13 clinical and demographic attributes, using a single-table CSV extracted to Hyper with no row limit, delivered as a fixed 1900x1050 dark-theme canvas and published to Tableau Public.
- Engineered the semantic layer entirely in Tableau: applied display captions to all 13 source fields, converted six binary integer flags from measures to ordinal dimensions, and defined data source level value aliases decoding 0/1 codes into clinical terms (Survive/Death, Positive/Negative, Male/Female, Smoker/Non-Smoker) so all 19 worksheets inherited the decoding from a single definition.
- Constructed five dual-axis donut charts from a constant-zero placeholder measure with a crossed two-field colour encoding and a Percent of Total table calculation computed along Rows, giving each comorbidity a simultaneous count and share view against the mortality outcome.

- Designed seven fixed-width binned histograms and two disaggregated views, including a dot plot with an **average** reference line scoped per cell, to expose distribution and spread of six clinical biomarkers and follow-up time by survival outcome.
- Implemented dashboard interactivity with a dashboard-scoped highlight action on the mortality outcome field and a Sex dropdown filter applied to a 14-worksheet filter group, built on a fully tiled nested-container layout with an auto-generated phone layout.

PART B

Hands-on evidence brief

Sections 1 to 13 for a hiring manager confirming hands-on experience: context, source inventory, source-to-target mapping, data flow, cleaning rules, data model, business rules and governance, operations and security, requirements to KPIs, stakeholder coordination, outcomes, open questions and a sanitised version with its substitution list.

Scoped to **Healthcare - Heart Failure** only. Structured for a hiring manager confirming hands-on experience.

Evidence base. The packaged workbook **Healthcare - Heart Failure.twb**, parsed from its XML, plus a short set of answers the author gave directly. The two are kept clearly apart throughout. Where neither supplies an answer, the entry reads "**Not recorded**" and the question appears in section 12.

Confirmed by the author, not by the file:

Item	Answer
Author	Atharva Devne
Why it was built	Self-directed, for the author's own skill development. Not a course, not a client, not an employer.
Dataset origin	Kaggle
Dataset public?	Yes. This resolves the only disclosure concern the file raised, since the workbook embeds a full extract and publishes to Tableau Public.

Scope note. This is an individual Tableau dashboard build over one local flat file. It is not pipeline, warehouse or platform work, and this brief does not pretend otherwise. Sections 2, 4, 6, 8 and 10 are therefore short, and the reasons are stated rather than padded.

1. PROJECT CONTEXT

Item	Detail
Business problem	No stated problem. The workbook contains no description, annotation, caption or text object of any kind. What the build addresses, read from its structure: every one of the 14 charts encodes <code>Death Event</code> on Colour and 12 chart titles end in <code>- Survival Stats</code> , so it is a survival profiling view. For each recorded clinical or demographic attribute, show how the cohort splits between <code>Survive</code> and <code>Death</code> .
Who requested it	Nobody. Self-directed.
What decisions it supports	No external decision. The purpose was the author's own skill development. What it mechanically enables a reader to ask is in Part A section 16.
My role	Sole author and sole builder. Every layer is one person's work: connection, extract, semantic layer, calculations, 19 worksheets, dashboard layout, filter scoping, action design, publication.
Who else was involved	Nobody. No collaborators, no reviewers. The workbook contains no second person: every data source is <code>inline='true'</code> (embedded, not published to a server) and every connection points at one local file on one machine.
Timeline	Not recorded.
Tool	Tableau Desktop, <code>source-build='2025.1.0 (20251.25.0313.2002)'</code> , <code>source-platform='mac'</code> , document format <code>version='18.1'</code> .
Current status	Published to Tableau Public. <code>repository-location</code> id <code>Healthcare-HeartFailure_17429983482570</code> , derived from <code>?rev=1.1</code> , current <code>revision='1.2'</code> , so at least two publish revisions. Whether it is live right now could not be checked: the network policy in the documenting environment denied <code>public.tableau.com</code> .

2. SOURCE SYSTEM AND DATA INVENTORY

There is no source system in the enterprise sense. No database, no API, no SharePoint, no shared folder, no server.
One local CSV on one Mac.

Attribute	Value
System	Local file system
Connector	<code>textscan</code> , Tableau's delimited text file connector
File	<code>heart_failure_clinical_records_dataset.csv</code>
Path	<code>/Users/atharvadevne/UIC/Business Data Visualization/Heart Fail Prediction</code>
Named connection id	<code>textscan.0eo7kao1bn80dv0zsqdco0t9bxql</code>
Parsing configuration	<code>character-set='UTF-8', separator=',', header='yes', locale='en_US'</code>
Relation	<code><relation name='heart_failure_clinical_records_dataset.csv' table='[heart_failure_clinical_records_dataset#csv]' type='table' /></code>
Schema	13 columns, flat, no key declared
Provenance	Kaggle, public dataset. Confirmed by the author.

Attribute	Value
Owner	Publicly available, no internal owner
Access method	Direct file read, then full extract to Hyper
Legacy versus current	Not applicable, one file, no history, no second system

Exact schema, with ordinals and declared types

Ordinal	Column	Declared type	Tableau remote-type
0	age	real	5
1	anaemia	integer	20
2	creatinine_phosphokinase	integer	20
3	diabetes	integer	20
4	ejection_fraction	integer	20
5	high_blood_pressure	integer	20
6	platelets	real	5
7	serum_creatinine	real	5
8	serum_sodium	integer	20
9	sex	integer	20
10	smoking	integer	20
11	time	integer	20
12	DEATH_EVENT	integer	20

Every metadata record carries `<contains-null>true</contains-null>` and the shared `object-id` `[heart_failure_clinical_records_dataset.csv_6F1533318D5F4C8AB71453465976CF29]`. The null flag is Tableau's default optimistic value on a scanned text file, **not** evidence that nulls exist.

All 13 columns are consumed. Nothing in this dataset was imported and then ignored.

3. SOURCE-TO-TARGET MAPPING

Target field	Source field	Transformation rule	Notes
Age	age	Caption rename only. Type <code>real</code> , role measure.	Used as <code>AVG</code> on the KPI, <code>CNT</code> on the histogram, unaggregated on the scatter
Death Event	DEATH_EVENT	Caption rename; role measure → dimension <code>ordinal</code> ; alias <code>0</code> → <code>Survive</code> , <code>1</code> → <code>Death</code>	The alias is literally <code>Survive</code> in the file, a typo for "Survive", and it renders that way in every legend and label

Target field	Source field	Transformation rule	Notes
Anaemia	anaemia	Caption rename; role → dimension <code>ordinal</code> ; alias <code>0</code> → <code>Negative</code> , <code>1</code> → <code>Positive</code>	
Diabetes	diabetes	Same pattern	
High Blood Pressure	high_blood_pressu re	Same pattern	
Sex	sex	Caption rename; role → dimension <code>ordinal</code> ; alias <code>0</code> → <code>Female</code> , <code>1</code> → <code>Male</code>	
Smoking	smoking	Caption rename; role → dimension <code>ordinal</code> ; alias <code>0</code> → <code>" Non-Smoker"</code> , <code>1</code> → <code>Smoker</code>	The Non-Smoker alias carries a leading space
Creatinine Phosphokinase	creatinine_phosph okinase	Caption rename only	
Ejection Fraction	ejection_fraction	Caption rename only	
Platelets	platelets	Caption rename only	
Serum Creatinine	serum_creatinine	Caption rename only	
Serum Sodium	serum_sodium	Caption rename only	
Time	time	Caption rename only	Integer follow-up period, not a date
Age (bin)	age	<code>class='bin', size='4.48', peg='0', decimals='0'</code>	
Creatinine Phosphokinase (bin)	creatinine_phosph okinase	<code>class='bin', size='376', peg='0', decimals='2'</code>	
Ejection Fraction (bin)	ejection_fraction	<code>class='bin', size='8.22', peg='0', decimals='0'</code>	
Platelets (bin)	platelets	<code>class='bin', size='41767', peg='0', decimals='4'</code>	
Serum Creatinine (bin)	serum_creatinine	<code>class='bin', size='0.772', peg='0', decimals='-1'</code>	
Serum Sodium (bin)	serum_sodium	<code>class='bin', size='3.57', peg='0', decimals='0'</code>	
Time (bin)	time	<code>class='bin', size='15.1', peg='0', decimals='1'</code>	
Total Individuals	none	<code>"Total Individuals"</code>	String literal, KPI caption
Total Deaths	none	<code>"Total Deaths"</code>	String literal, KPI caption
Total Males	none	<code>"Total Males"</code>	String literal, KPI caption
Total Females	none	<code>"Total Females"</code>	String literal, KPI caption
Average Age	none	<code>"Average Age"</code>	String literal, KPI caption
0	none	<code>0</code>	Dual-axis placeholder measure

Fields whose meaning differed between systems. Not applicable. One system, so no cross-system semantic mismatch existed and none was resolved.

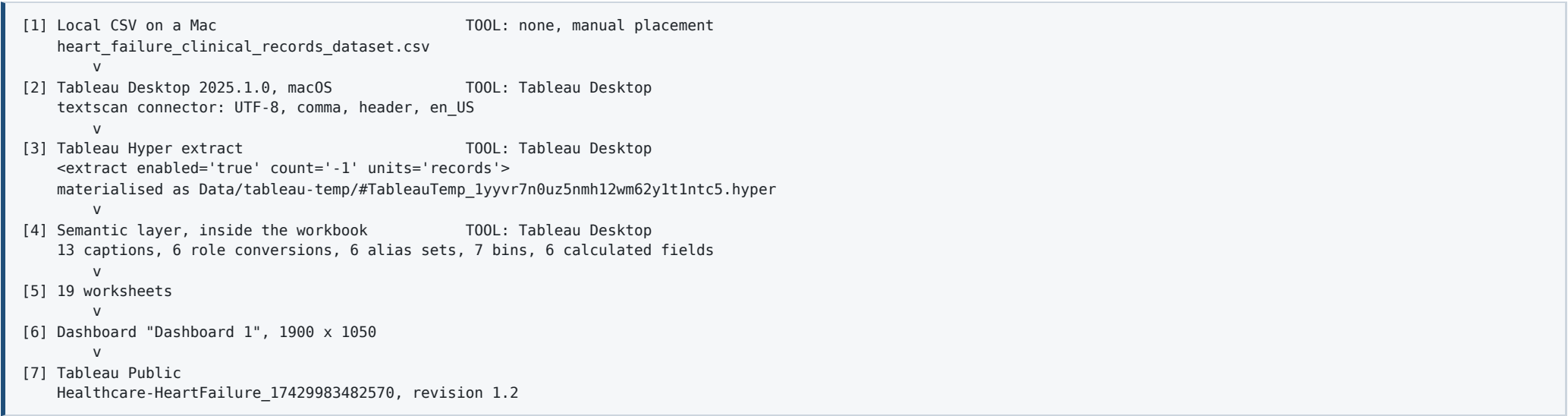
Grain

Object	Grain
Source	One row per patient. Inferred from the schema: no patient identifier, no repeated-measure column, no date, and every column is a single clinical attribute.
KPI sheets	Whole cohort, no dimensions
Donuts	One mark per Death Event x comorbidity combination
Histograms	One mark per bin x Death Event
Sheet 6 (8) scatter	One mark per distinct age x time combination
Sheet 6 (9) dot plot	Record grain , because aggregation='false'

Grain mismatches: none. Every target aggregates upward from a single-grain source. No fan-out, no many-to-many, no double-counting risk, because there is no join anywhere.

4. DATA FLOW AND ARCHITECTURE

No orchestration tool, no staging layer, no warehouse, no lakehouse, no scheduler.



Stage	Where data lands
Source	Local file system

Stage	Where data lands
Staging	None
Warehouse / lakehouse	None
Curated files	The Hyper extract inside the packaged workbook. The only materialised intermediate.
Serving	Tableau Public

Inherited or previous architecture: none. Nothing in the workbook records a predecessor, a migration or a rebuild.

Flow in boxes and arrows: local CSV → Tableau Desktop textscan connector → Hyper extract (all records, no filter) → semantic layer (captions, roles, aliases, bins) → 19 worksheets → 1 dashboard → Tableau Public. A side box, "filter scoping and one highlight action", feeds into the dashboard box. There are no other boxes: no SQL step, no Python step, no Power Query, no Alteryx, no Airflow, no dbt.

5. DATA CONSOLIDATION, CLEANING AND STANDARDISATION

No consolidation occurred. One table in, one table used.

Every cleaning rule applied, all four

#	Rule	Exact definition	Fields affected	Layer	Why there
1	Display renaming	<code>caption</code> attribute on <code><column></code>	All 13 source columns	Data source	Defined once, inherited by all 19 worksheets
2	Role reclassification	<code>role='dimension'</code> <code>type='ordinal'</code>	<code>DEATH_EVENT</code> , <code>anaemia</code> , <code>diabetes</code> , <code>high_blood_pressure</code> , <code>sex</code> , <code>smoking</code>	Data source	Tableau imports 0/1 integers as measures and defaults to <code>SUM</code> , which is meaningless for a flag. Dimensions can drive Colour, Label and Filter.
3	Code-to-label standardisation	<code><aliases_enabled='yes'></code> with explicit key/value pairs	6 fields, 12 aliases	Data source	Same reason: one definition, 19 inheritors
4	Binning	<code><calculation class='bin' ... /></code>	7 continuous measures	Data source	Required to draw a distribution of a continuous measure

Exact alias mappings

Field	Key	Aliased value
<code>DEATH_EVENT</code>	0	<code>Survive</code>
<code>DEATH_EVENT</code>	1	<code>Death</code>
<code>anaemia</code>	0 / 1	<code>Negative</code> / <code>Positive</code>
<code>diabetes</code>	0 / 1	<code>Negative</code> / <code>Positive</code>

Field	Key	Aliased value
high_blood_pressure	0 / 1	Negative / Positive
sex	0 / 1	Female / Male
smoking	0 / 1	" Non-Smoker" (leading space) / Smoker

Cleaning rules NOT applied

Stated so nothing is over-claimed.

Rule category	Status
Null handling (ISNULL , IFNULL , ZN)	Not present
Duplicate detection or removal	Not present
Explicit type casting (INT() , STR() , FLOAT() , DATE())	Not present. All types came from the CSV scan.
Date parsing, date parts, timezone handling	Not present, and not possible. The dataset has no date column. time is an integer follow-up period.
Text normalisation (TRIM , UPPER , REPLACE , REGEXP)	Not present. Note that the workbook's own " Non-Smoker" alias is itself untrimmed.
Outlier detection or capping	Not present
Country, currency or unit standardisation	Not present. No such field exists.
Domain classification logic	Not present
Referential validation	Not present. No joins to validate.
Data source filter	Not present
Extract filter	Not present
Tableau Prep flow	Not present in the archive
Groups, sets, hierarchies	Not present

Which layer performed each transformation, and why there

All of it was done inside Tableau Desktop. No SQL, no Python, no Prep flow is referenced anywhere. Whether anything was cleaned before Tableau is **not recorded**.

Within Tableau, a deliberate layering is visible:

Layer	What sits there	Why
Data source scope	Captions, role conversions, aliases, bins	Defined once, inherited everywhere. Changing one alias updates 19 sheets.
Calculated field scope	7 bins, 5 string literals, the 0 placeholder	Reusable across sheets
Worksheet scope	The three hard-coded KPI member filters	Local to one card, no reason to promote

Layer	What sits there	Why
Dashboard scope	Sex filter control, colour legend, highlight action	Interaction, not data

6. DATA MODEL: SCHEMA AND TABLE DESIGN

No dimensional model was designed. This is the single most important scope boundary to understand about this artefact.

Design element	Status
Fact tables	None
Dimension tables	None
Star or snowflake schema	None
Surrogate key strategy	None. No key column is declared or used.
Slowly changing dimensions	None
Date dimension	None, and impossible. No date column exists.
Relationships and filter directions	None. One table.
Historical migration or backfill	None
Cutover between systems	Not applicable
Cardinality and referential integrity settings	Not applicable

Design rules that WERE set, and the reason for each

Real, verifiable structural decisions, even though they are not dimensional modelling decisions.

#	Rule	Evidence	Reason
1	Extract everything, filter nothing at extract time	<code>count='-1'</code> , no extract filter	Keeps the workbook self-contained and keeps all filtering interactive
2	Embed the data source rather than publish it	<code>inline='true'</code>	Self-contained portfolio artefact
3	Binary flags are dimensions, not measures	<code>role='dimension' type='ordinal'</code> on 6 fields	Prevents meaningless <code>SUM</code> on a flag; enables Colour, Label, Filter
4	Decode codes at the source, not in the view	<code><aliases enabled='yes'></code> at data source level	One definition, 19 inheritors
5	Scope a filter to a chosen subset of sheets	<code>filter-group='3'</code> on 14 of 19 sheets	Applies one control to the charts. Consequence: the 5 KPI cards are excluded.
6	One visual grammar, repeated	5 structurally identical donuts, 7 structurally identical histograms	The reader learns the chart once and reads six more for free
7	Hide axes that carry no information	<code>major-show='false'</code> , blank axis titles on the donut sheets	The constant <code>0</code> axis exists only to enable the dual axis

7. BUSINESS RULES, VALIDATION AND GOVERNANCE

Business rules encoded

None. This workbook encodes no business rule at all. Its aliases are label mappings, not rules. There is no classification logic, no exclusion, no threshold, no derived category and no conditional expression anywhere in it. The only decision logic is the three hard-coded member filters on the KPI cards (`DEATH_EVENT = 1` , `sex = 1` , `sex = 0`), and those are filters, not rules.

Validation checks

None are present. A complete absence, not an omission from this brief.

Check type	Present?
Reconciliation to source	No. No reconciliation sheet, control total or comparison view.
Control totals	No
Cross-system checks	Not applicable, one system
Step-change or regression checks	No
Data-quality flags shown to users	No. No worksheet, annotation or field surfaces a quality indicator.

Whether validation was performed outside Tableau is **not recorded**.

Governance

Item	Status
Metric definitions and data dictionary	None. No documentation field, description or dictionary.
Sign-off process and owners	None. Self-directed work, nothing to sign off.
Decision log	None
Open questions register	None
Documentation	None in the workbook
Version control	Tableau Public revision counters only: <code>revision='1.2'</code> , derived from <code>?rev=1.1</code> , so at least two publishes

Defects observable in the delivered artefact

Found by reading the XML, not reported by anyone. All six are **unfixed in the delivered file**.

#	Defect	Evidence
1	The <code>DEATH_EVENT = 0</code> alias is spelled <code>Survive</code> , a misspelling of "Survive", and it renders in every legend and label	<code><alias key='0' value='Survive' /></code>
2	The <code>smoking = 0</code> alias has a leading space : <code>" Non-Smoker"</code>	<code><alias key='0' value=' Non-Smoker' /></code>
3	The Sex dashboard filter is scoped to <code>filter-group='3'</code> , covering 14 of 19 sheets and excluding all five KPI cards , so the headline numbers do not move when the filter is used	Filter group membership per worksheet
4	All 7 bin sizes are non-round values (<code>4.48</code> , <code>8.22</code> , <code>0.772</code> , <code>3.57</code> , <code>15.1</code> , <code>376</code> , <code>41767</code>), consistent with Tableau's suggested defaults rather than chosen widths	Bin <code>size</code> attributes
5	No number formatting is defined anywhere	No <code>number-format</code> attribute in the file
6	No custom tooltip on any of the 19 worksheets	No <code><customized-tooltip></code> element in the file

A seventh point, not a defect but a design consequence worth being able to defend: `Total Individuals` uses `CNT([DEATH_EVENT])` rather than `COUNT(*)` or an explicit `Number of Records`. It counts non-null values, so it equals the cohort size only if `DEATH_EVENT` is never null.

8. REFRESH, OPERATIONS AND SECURITY

Item	Status
Refresh frequency	Not recorded. No schedule, no incremental configuration, no refresh definition exists.
Full or incremental	Full, by construction. <code>count='-1' units='records'</code> , no incremental refresh key, no extract filter. An extract configured this way refreshes in full when refreshed at all.
Refresh trigger	The connection is to a local file, so a refresh requires that file to be present on that Mac and a manual refresh to be run.
How the source connects to Tableau	Direct local file read via the <code>textscan</code> connector, then extract. No server, no gateway, no DSN, no connection string, no credentials, no DirectQuery equivalent. The live-versus-extract choice is extract .
Connection parameters	Text parsing options only: <code>UTF-8</code> , <code>,</code> , header row, <code>en_US</code>
Credentials handling	<code>workgroup-auth-mode='as-is'</code> , the default for a file connection needing no authentication. No credential is stored, because none is needed.
Access control	Published to Tableau Public , which is public by default with no user-based access control.
Row-level security	None. No user filter, no <code>USERNAME()</code> , no <code>ISMEMBEROF()</code> , no user function appears anywhere.
Workspace administration	Not applicable. Tableau Public has no workspace model.
What data was allowed to leave the source	No restriction, and none needed. The workbook is packaged with a full data extract embedded, so publishing it publishes the underlying records, downloadable by any viewer. The dataset is a public Kaggle dataset , confirmed by the author, so there is no disclosure issue. Worth stating explicitly: embedding a full extract in a public workbook would be a real problem with any non-public source, and knowing that distinction is the point.

Item	Status
Monitoring, failure handling, notification	None. No alert, subscription or notification is configured.

9. REQUIREMENTS TO KPIs AND DASHBOARD LOGIC

How requirements were gathered

No requirements were gathered. Self-directed work with no stakeholder, no brief and no session. Nothing is recorded in the workbook either.

KPI list

KPI	Business definition	Decision it supports	Grain	Source fields	Calculation logic	Owner	Sign-off
Total Individuals	Count of patients in the cohort	Sets the base for reading every other figure	Whole cohort, unfiltered	DEATH_EVENT	CNT([DEATH_EVENT]) , no filter	Author	None, self-directed
Total Deaths	Count of patients with the mortality flag set	Absolute mortality volume	Filtered to DEATH_EVENT = 1	DEATH_EVENT	CNT([DEATH_EVENT]) with worksheet filter member='1'	Author	None
Total Males	Count of male patients	Cohort sex composition	Filtered to sex = 1	sex	CNT([sex]) with worksheet filter member='1'	Author	None
Total Females	Count of female patients	Cohort sex composition	Filtered to sex = 0	sex	CNT([sex]) with worksheet filter member='0'	Author	None
Average Age	Mean patient age	Frames the age distribution below it	Whole cohort, unfiltered	age	AVG([age])	Author	None

None of the five responds to the dashboard's Sex filter, because the five KPI worksheets sit outside filter-group='3' .

There is no rate KPI. Mortality rate is never computed as a KPI, only counts and one mean.

Measures written, actual code

No DAX and no SQL exists. The Tableau code, transcribed exactly:

```
// The five KPI caption fields
"Total Individuals"
"Total Deaths"
"Total Males"
"Total Females"
"Average Age"

// Dual-axis placeholder
0

// The seven bin definitions
class = bin, formula = [age], size = 4.48, peg = 0, decimals = 0
class = bin, formula = [creatinine_phosphokinase], size = 376, peg = 0, decimals = 2
class = bin, formula = [ejection_fraction], size = 8.22, peg = 0, decimals = 0
class = bin, formula = [platelets], size = 41767, peg = 0, decimals = 4
class = bin, formula = [serum_creatinine], size = 0.772, peg = 0, decimals = -1
class = bin, formula = [serum_sodium], size = 3.57, peg = 0, decimals = 0
class = bin, formula = [time], size = 15.1, peg = 0, decimals = 1

// The one table calculation, applied on 5 sheets
<table-calc ordering-type='Rows' type='PctTotal' />
```

There are no LOD expressions. Verified by absence of `{` in every formula.

Dashboard design

Item	Value
Pages	1 dashboard, no story, no navigation
What it shows	Top strip: 5 KPI cards. Second strip: 5 donuts, one per binary attribute, crossed with the survival outcome. Third and fourth strips: 7 biomarker histograms plus 2 scatter-style views.
Visuals used	Donut built as a dual-axis pie, stacked histogram, scatter, disaggregated dot plot with a per-cell average reference line, text cards
Filters shown to the user	1 dropdown, <code>Sex</code> , applied to 14 of 19 sheets
Parameter controls	None. There are no parameters.
Drill paths	None. No hierarchy exists.
How assumptions and definitions are shown to the user	They are not. No text object, no annotation, no caption, no custom tooltip anywhere on this dashboard.
Canvas	1900 x 1050, <code>sizing-mode='range'</code> , black, fully tiled with nested flow containers

Anything the business explicitly asked for in the presentation: nothing. There was no business.

10. STAKEHOLDER COORDINATION

Nothing to report, and that is accurate rather than a gap in the evidence. The project was solo and self-directed. There were no stakeholders, no meetings, no working sessions, no agreements, no pushback and no disagreements to handle. The workbook contains no name, comment, review or approval artefact of any kind.

11. OUTCOMES AND STATUS

Item	Status
What is live	Published to Tableau Public, workbook id <code>Healthcare-HeartFailure_17429983482570</code> , dashboard repository id <code>Dashboard1</code> . Published at least twice (revision advanced <code>1.1</code> → <code>1.2</code>). Whether currently live could not be checked, the documenting environment's network policy denied <code>public.tableau.com</code> .
Completeness	Complete build. All 19 worksheets in the workbook are placed on the dashboard. No orphaned, unfinished or staged worksheets.
What is validated	Nothing. No validation artefact exists.
What is signed off	Not applicable, self-directed.
What is still open	The 6 defects in section 7, all unfixed. The three most consequential: the <code>Survive</code> misspelling, the KPI cards being outside the filter scope, and the absence of any custom tooltip across 19 worksheets.

Measurable results

None can be claimed. No view count, adoption figure, time saving, grade, feedback or business outcome exists. No data values were read either, because the `.hyper` extract is binary and no Hyper reader was available, so no mortality rate, distribution shape or correlation can be stated.

What can be counted, and only this:

Metric	Value
Worksheets built	19
Worksheets placed on the dashboard	19 of 19
Calculated fields authored	13
Source columns consumed	13 of 13
Distinct chart forms	5
Dual-axis constructions	5

Metric	Value
Table calculations	1 type, on 5 sheets
Reference lines	1
Value aliases defined	12, across 6 fields
Dashboard actions	1
Custom worksheet titles	14
Publish revisions	at least 2

Lessons learned

Not recorded in the workbook. The defect list in section 7 is an external reading of the file, not a record of anything the author wrote down.

12. GAPS AND QUESTIONS

Already answered

Question	Answer
Was this for a course, a client, or self-directed?	Self-directed, for your own development. The <code>UIC/Business Data Visualization</code> folder is just where the file sat.
Where did the dataset come from?	Kaggle.
Is the dataset public, given the embedded extract?	Yes, public. No disclosure issue.

Still open

1. **Was the CSV cleaned before it reached Tableau?** If yes, with what tool and what exact rules?
2. **Does `DEATH_EVENT` contain any nulls?** `CNT([DEATH_EVENT])` is the cohort-size KPI and it counts non-nulls, so nulls would silently understate it.
3. **Why counts and no mortality rate?** Was a rate considered and rejected?
4. **Why were Tableau's suggested bin sizes accepted** (`4.48` , `8.22` , `0.772` , `3.57` , `15.1`)? Was a clinically meaningful width considered?
5. **Was excluding the KPI cards from the Sex filter deliberate**, so the cards act as a constant baseline, or an oversight?
6. **Why a highlight action rather than filter actions?** This is a genuinely good design question and a strong answer lands well.

7. **Why no custom tooltips across 19 worksheets?** Deliberate minimalism on a dark theme, or not reached?
 8. **Did anyone review or use it**, and what did they say?
 9. **The confirmed live Tableau Public URL.** The recoverable id is `Healthcare-HeartFailure_17429983482570`.
 10. **Any view counts or feedback?**
 11. **What changed between the two publish revisions?**
 12. **One concrete insight you personally found** in this dataset while building. Not a figure the dashboard computes, but something you noticed. Interviewers ask "what did you find", and "the dashboard lets you find X" is a weaker answer than "I found X".
 13. **Which of the 6 defects do you agree with**, and does any have an explanation this reading has missed?
-

13. SANITISED VERSION

The same brief with identifying details replaced. All technical detail, counts, rules and logic unchanged.

1. Project context. Self-directed work by a single author for their own skill development, not for a course, client or employer. No requester, no stakeholder, no collaborators. Subject matter inferable from structure: profiling patient survival in a cardiac care dataset, evidenced by 12 of 14 chart titles ending in a survival-statistics suffix and a mortality outcome field on the Colour shelf of every chart. The workbook contains no description, annotation, caption or text object. BI tool version 2025.1.0 on a workstation. Published to a public visualisation hosting platform, at least twice.

2. Source system and data inventory. No enterprise source system. One local delimited file on one workstation, read through the delimited text connector with `character-set='UTF-8'`, `separator=','`, `header='yes'`, `locale='en_US'`, one relation of `type='table'`, 13 columns, no key declared. Provenance: a public dataset from a public data-sharing platform. Schema: patient age (real), four binary comorbidity or lifestyle flags (integer), five continuous biomarker measurements (three real, two integer), a binary demographic flag (integer), an integer follow-up period, and a binary mortality outcome (integer). All 13 columns are consumed. Every metadata record carries the connector's default optimistic null flag, which is not evidence that nulls exist.

3. Source-to-target mapping. Thirteen source columns mapped as: caption rename on all 13; role conversion measure → dimension `ordinal` on six binary flags; value aliases decoding `0` and `1` into readable terms on those same six, including a misspelled survival label and an untrimmed leading space on one lifestyle label; seven bin fields with `peg='0'` and fixed sizes `4.48`, `376`, `8.22`, `41767`, `0.772`, `3.57`, `15.1` and `decimals 0, 2, 0, 4, -1, 0, 1`; five string-literal caption fields; and one constant `0` dual-axis placeholder. No cross-system semantic mismatch existed, because there is one system. Source grain is one row per patient; target grains range from whole-cohort to record grain on the one worksheet with aggregation switched off. No grain mismatches, because no join exists.

4. Data flow. Local delimited file → BI desktop tool, delimited text connector → columnar in-memory extract, all records, no filter → semantic layer (captions, roles, aliases, bins) → 19 worksheets → one dashboard → public hosting platform. No staging, no warehouse, no lakehouse, no scheduler, no SQL step, no scripting step, no ETL tool. No inherited architecture.

5. Cleaning and standardisation. No consolidation, one table. Four rules: display renaming on all 13 columns at data source scope; role reclassification on six binary integer columns because the tool imports them as measures and defaults to summing a flag; code-to-label standardisation via data source aliases, six fields and twelve pairs; and binning of seven continuous measures. Not applied: null handling, duplicate detection, explicit type casting, date parsing or timezone handling (impossible, no date column), text normalisation, outlier treatment, unit or code standardisation, referential validation, data source filter, extract filter, external preparation flow, groups, sets, hierarchies. All work was done inside the BI tool across four deliberate layers: data source scope for anything inherited everywhere, calculated field scope for reusable definitions, worksheet scope for view-local devices, dashboard scope for interaction.

6. Data model. No dimensional model. No fact or dimension tables, no surrogate keys, no slowly changing dimensions, no date dimension (impossible, no date column), no relationships, no migration, no cutover. Seven design rules were set and are listed with reasons: extract everything and filter nothing at extract time; embed rather than publish the data source; treat binary flags as dimensions; decode codes at the source; scope a filter to a chosen subset of sheets; repeat one visual grammar; hide axes that carry no information.

7. Business rules, validation, governance. No business rule is encoded at all; the aliases are label mappings, not rules. No validation checks of any kind: no reconciliation, control total, step-change check, regression check or data-quality flag. No data dictionary, sign-off, decision log or documentation. Version control limited to publish revision counters. Six defects are verifiable in the delivered file, all unfixed: a misspelled outcome label that renders in every legend; an untrimmed leading space in one lifestyle label; a dashboard filter scoped to a sheet group that excludes all five KPI cards; seven bin sizes matching the tool's suggested defaults rather than chosen widths; no number formatting anywhere; and no custom tooltip on any of the 19 worksheets.

8. Refresh, operations, security. Refresh frequency not recorded. Full by construction: all records, no incremental key, no extract filter. Direct local file read then extract, with no server, gateway, DSN, connection string, credentials or direct-query equivalent. Authentication mode is the default for a file connection needing none, so no credential is stored. Published to a public-by-default platform with no user-based access control and no row-level security; no user function appears anywhere. The workbook embeds a full extract, so publishing it publishes the records, but the dataset is public so there is no disclosure issue. No monitoring, failure handling or notification.

9. Requirements to KPIs. No requirements were gathered; the work was self-directed. Five KPIs: cohort size as a count of non-null outcome values with no filter; a death count as the same count filtered to the outcome member **1**; two demographic counts each filtered to one member of the demographic flag; and a mean age. None of the five responds to the dashboard filter. There is no rate KPI. No DAX and no SQL; the actual code is five string literals, one zero constant, seven bin definitions and one built-in percent-of-total table calculation, all transcribed unchanged in the full section 9. No level-of-detail

expressions. Dashboard: one page, a KPI strip of five cards, a strip of five donuts crossing each binary attribute with the outcome, then seven biomarker histograms and two scatter-style views, on a 1900 x 1050 range-sized dark canvas, fully tiled with nested flow containers. One dropdown filter on 14 of 19 sheets, no parameter controls, no drill paths. Assumptions and definitions are not shown to the user at all.

10. Stakeholder coordination. None. Solo, self-directed work with no stakeholders, meetings, agreements or disagreements.

11. Outcomes and status. Published at least twice to a public hosting platform. Complete build: all 19 worksheets placed on the dashboard, none orphaned. Nothing validated, nothing signed off, six defects still open. No measurable results can be claimed and no data values were read. Countable facts: 19 worksheets, 13 calculated fields, 13 of 13 source columns consumed, 5 distinct chart forms, 5 dual-axis constructions, 1 table calculation type on 5 sheets, 1 reference line, 12 value aliases, 1 dashboard action, 14 custom titles, at least 2 publish revisions.

12. Gaps and questions. Three answered (self-directed, public data-sharing platform origin, dataset is public). Thirteen still open, listed in full in the unsanitised section 12, covering pre-tool cleaning, nulls in the outcome column, why counts and not a rate, why default bin widths, whether the KPI filter exclusion was deliberate, why a highlight action, why no tooltips, review and feedback, the confirmed live URL, view counts, what changed between revisions, one concrete personal finding, and which defects are agreed.

Substitution list

#	Original	Replaced with
1	Tableau, Tableau Desktop, Tableau Public	"BI desktop tool", "public visualisation hosting platform"
2	Hyper, .hyper	"columnar in-memory extract"
3	textscan connector	"delimited text connector"
4	Atharva Devne	"a single author" / "the author"
5	atharvadevne (account name in file paths)	omitted
6	/Users/atharvadevne/UIC/Business Data Visualization/Heart Fail Prediction	omitted
7	UIC	omitted
8	Kaggle	"a public data-sharing platform"
9	Healthcare - Heart Failure.twb	"the workbook"
10	heart_failure_clinical_records_dataset.csv	"one local delimited file"
11	Healthcare-HeartFailure_17429983482570	omitted
12	public.tableau.com	"the host"
13	Dashboard 1, Dashboard1	"the dashboard"

#	Original	Replaced with
14	anaemia, diabetes, high_blood_pressure	"binary comorbidity flags"
15	smoking	"lifestyle flag"
16	sex	"binary demographic flag"
17	DEATH_EVENT	"mortality outcome"
18	Survive / Death	"a misspelled survival label"
19	" Non-Smoker"	"an untrimmed leading space on one lifestyle label"
20	creatinine_phosphokinase, ejection_fraction, platelets, serum_creatinine, serum_sodium	"five continuous biomarker measurements"
21	time	"integer follow-up period"
22	age	"patient age"
23	Worksheet names (Sheet 1, Sheet 6 (9) , and so on)	described by chart type
24	Federated data source id federated.15n4u4511e3xfk1749m9r1i714e1	omitted
25	Internal calculation ids ([Calculation_215609893383794688] and so on)	replaced with the readable field name
26	Dashboard UUID {14818E80-7948-4984-90AD-D1E566C79BE9}	omitted
27	Extract filename #TableauTemp_1yyvr7n0uz5nmh12wm62y1t1ntc5.hyper	omitted
28	macOS	"workstation"

Not substituted, deliberately: all bin sizes, decimal settings, formulas, colour hex codes, canvas dimensions, zone counts, filter scopes, action counts, worksheet counts and column counts, because the instruction was to keep technical detail, counts, rules and logic unchanged.

One item to check yourself: the sanitised version still states that the subject is cardiac patient survival. If the subject matter itself is sensitive, replace it with something like "a clinical outcome dataset".